

The Dynamic Consequences of State Building: Evidence from the French Revolution[†]

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How do radical reforms shape economic development over time? In 1790, the French Constituent Assembly overhauled the kingdom's organization to establish new local capitals. In some departments, the choice of local capitals over rival candidate cities was plausibly exogenous. We study how changes in administrative presence affect state capacity and development in the ensuing decades. In the short run, administrative proximity increases taxation and investments in law enforcement. In the long run, capitals obtain more public goods and grow faster. Our results shed light on the dynamic impacts of state building following one of history's most ambitious administrative reforms. (JEL D70, H41, H71, O18, O43)

State capacity is considered essential for economic development.¹ History, however, provides many examples of both failures and successes of state building. Success seems to depend on how the different dimensions of state capacity evolve and expand over time. The literature has highlighted the importance of early investments in the state's capacity to enforce law and order and to collect resources from citizens (Besley and Persson 2010). However, there is little evidence on the dynamic evolution of these and other functions of the state, such as public goods provision, and its consequences for economic growth.

We exploit one of history's most ambitious state-building experiments, the administrative reform initiated by the French Revolution, to explore the dynamics of

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¹ In economics, Acemoglu, García-Jimeno, and Robinson (2015); Becker et al. (2020); Cantoni, Mohr, and Weigand (2024); Dell, Lane, and Querubin (2018); Dinzeo and Katz (2014); Gennaioli and Voth (2015); and Johnson and Koyama (2017), among many others, provide recent empirical evidence.

state building and its economic impacts. In February 1790, France's first Constituent Assembly overhauled the administration of the kingdom to set up a radically different organization of the state. In response to widespread grievances about administrative complexity before the Revolution, the Assembly established new centers concentrating local administrative functions. The main achievement of the 1790 reform was to divide the territory into homogeneous units, known as *départements*, in which citizens would never be more than a day of travel away from their nearest local capital. After 1800, Napoléon solidified this reform and gave additional relevance to these new administrative centers by appointing civil servants and by creating local state institutions, which largely survived to the present day.

Our identification strategy exploits a unique feature of the 1790 reform. Facing pressure to design a reform in a few months, the Assembly silenced competing local demands in a number of artificial new departments by establishing a rotation (*alternat*) of state functions across multiple cities or by allowing the capital to be chosen from a set of candidate cities through a local election. The Assembly eventually abolished rotations and ordered local administrations to remain where they were located in 1791; after continued uncertainty during the revolutionary period, France's administrative network remained nearly unchanged after 1800. We leverage this setting to study how plausibly exogenous variation in local administrative presence affects the state's coercive power, public goods provision, and economic development. We identify causal effects of state building by comparing capitals with candidate cities that were not ultimately chosen, and we also compare municipalities in the periphery of both types of cities. In support of our strategy, we show that final capitals were indistinguishable from other candidate cities along a wide range of pre-reform demographic and administrative characteristics.

A second key novelty of our paper is to build comprehensive data at a fine level both in space, at the municipal level, and in time, with a particular focus on the early nineteenth century. Several data sources allow us to explore the dynamic effects of state building. We estimate short-term effects of the reform on the establishment of municipal cadastres (land registries), military conscription, and the construction of prisons and tribunals. We then explore how the reform affected local public goods, including schools, hospitals, and public infrastructure such as railways. To measure impacts on economic activity, we digitized municipal-level data on the location of industrial establishments in 1839 and price data from 1849 and 1854; we also obtained patent registration and local banking activity data spanning the nineteenth century. Finally, we study how the reform affected population growth and employment in the long run. Together, these data sources allow us to characterize, for the first time, the short- and long-term economic impacts of the landmark 1790 reform at the level of cities and villages.

In our empirical analysis, we first study the dynamic evolution of extractive capacity—the state's ability to obtain fiscal or physical contributions from citizens. We use state-of-the-art methods in the randomization inference literature (DiCiccio and Romano 2017; Young 2024) to compare capitals and other candidates along a range of dimensions. Capitals and their periphery receive their first cadastre earlier, and they pay more business tax in the country's first industrial census conducted between 1839 and 1847. We also observe a rapid increase in the state's ability to enforce law and order. Capitals have more prisons and tribunals as early as 1815,

and they have a larger police force by 1816. Overall, the gap in coercive capacity between the capital and other candidate cities grows very early in the nineteenth century. The impacts of the reform on public goods and development, however, materialize later in the period; capitals initially do not experience differential investments in hospitals or secondary schools, with this divergence in public goods provision occurring in the second half of the nineteenth century.²

While these gradual investments in local public goods might have ultimately contributed to subsequent urban growth, the reform did not immediately stimulate economic development. Measures of productivity, innovation, the presence of banks, and prices and wages remain comparable between the capital and other candidate cities for several decades after the reform. Economic benefits only materialize over time, as a result of the comparative advantage induced by superior administrative capacity and public goods provision in the capital. From 1860 onward, capitals have greater access to the telegraph and railway networks, and they provide more welfare services. More banks operate in capitals and more patents are registered by their residents starting around 1870. Effects on population growth also become substantial by the turn of the twentieth century. By 1914, capitals are approximately 40 percent more populated than other candidate cities in 1790, with civil servants and their families explaining about a third of this difference. This divergence continues throughout the twentieth century; by 1999, department capitals are considerably more populated and attract substantially more public and private employment.

The growth benefits of state building also extend to towns and villages located in the periphery of capitals, but these benefits are modest and only materialize over the long run. To estimate these results, we compare municipalities whose nearest candidate city becomes the capital with those whose nearest candidate city was not chosen. Similar to the capitals themselves, municipalities in their immediate periphery are much more likely to receive their cadastre by 1815. Economic spillovers from the new capitals mainly pertain to innovation; by the end of the nineteenth century, the residents of municipalities close to capitals register significantly more patents. Two hundred years after the reform, these municipalities are 9 percent to 20 percent more populated than the municipalities in the vicinity of other candidate cities.

We present a simple spatial equilibrium model that delivers predictions consistent with these empirical results and shows that the early buildup of coercive capacity may be key to explain the dynamics of public goods provision and productivity. In this model, a central government chooses the tax rate and revenue allocation (between local public goods, global public goods, and investments in coercive capacity) across two cities. Private sector productivity and wages increase in the level of local public goods. Initially, the capital receives an exogenous shock in coercive capacity, allowing it to raise more taxes and to fund global public goods. These taxes drive citizens away, and the gap in coercive capacity grows even larger in favor of the capital. In the long run, the capital not only raises much more tax revenue due to its higher fiscal capacity

²In line with previous work, we conceptualize state capacity as the power of the state to “achieve intended outcomes” (Berwick and Christia 2018; Lindvall and Teorell 2017; Mann 1984). Throughout the paper, we categorize state functions into three types: extractive capacity, enforcement capacity, and productive capacity, that is, the ability to produce local and global public goods. Consistent with Besley and Persson (2010), we conceptualize extractive and enforcement capacity as inputs in the production of public goods, and we refer to the combination of these inputs as the coercive capacity of the state.

but also funds local public goods. Citizens are attracted by the higher productivity and supply of local public goods in the capital, which experiences population growth.

Our results provide insights on the recent successes and failures of state building. History contains many examples of successful states that first built up coercive capacity before they could invest in productive capacity. In the early days of the Roman Republic, in Europe's medieval communes, or in early Ming dynasty China (Szonyi 2017), mandatory military service was a key dimension of state power and often a prerequisite for participation in public life. By contrast, fragile states in recent years have faced numerous challenges in establishing coercive capacity. Recent work on the origins of state capacity in low-income settings tends to focus on fiscal capacity (Bergeron, Tourek, and Weigel 2021; Weigel 2020) in contexts where the state lacks the ability to enforce law and order (Sánchez de la Sierra 2020). France's experience, analyzed through the lens of postrevolutionary state building, illustrates that coercive capacity and public service provision must likely go hand in hand for strong states to emerge in the long run.

We relate to a large literature on state capacity. A first branch of this literature studies the long-term determinants (e.g., Besley and Persson 2010; Tilly 1990) and consequences (e.g., Acemoglu, García-Jimeno, and Robinson 2015; Dincecco and Katz 2014) of investments in state capacity, broadly defined. In related work, Bai and Jia (2023) show how changes in political hierarchy across Chinese cities affected population density and urbanization. Campante and Do (2014) study how the location of US state capitals affects government accountability. To our knowledge, our paper is the first in this literature to document how investments in state capacity unfold after a radical administrative reform in the short, medium, and long run. We identify which dimensions of state capacity are most responsive to policy efforts to create a modern state. The unique historical experiment associated with the 1790 reform also allows us to build an original identification strategy exploiting the plausibly exogenous allocation of administrative status.

Other papers in the literature emphasize specific dimensions of state building. Gennaioli and Voth (2015) show that fiscal capacity is triggered by conflict and responds to technological leaps in warfare. Cantoni, Mohr, and Weigand (2019) study the rise of fiscal capacity in the Holy Roman Empire and its consequences for the survival of local polities. Dittmar and Meisenzahl (2019) focus on the long-term productive benefits of legal innovations that spurred human capital investments in German cities during the 1500s. Johnson (2015) studies the relationship between fiscal capacity and nation building in postrevolutionary France. We unbundle these different dimensions of state capacity and show that early investments in coercive capacity yield economic payoffs in the long run. While previous work has studied the long-term consequences of investments in state capacity, one of our key contributions is to explore, conceptually and empirically, all the dynamics linking the short and the long run.

A related literature discusses the impact of public employment on local private sector outcomes. Several studies look at the impact of public sector relocation programs, such as those associated with the transfer of the German federal government (Becker, Heblich, and Sturm 2021; Faggio, Schluter, and vom Berge 2019; Faggio 2019). This literature provides mixed evidence on the extent of complementarities

between public and private employment. Public sector expansions tend to have a positive effect on private sector employment, particularly in the nontradable sector (Becker, Heblich, and Sturm 2021; Faggio, Schluter, and vom Berge 2019; Faggio 2019; Jofre-Monseny, Silva, and Vázquez-Grenno 2020; Faggio and Overman 2014; Guillouzouic, Henry, and Monras 2021), while Auricchio et al. (2020) finds that the effect does not hold for contractions. We depart in several ways from this literature. While these papers examine shocks in public employment in contexts where the state is already well established, we study instead the process of state creation before the advent of modern taxation and law enforcement systems and focus on the dynamic effects of public investments (both in labor and capital). Moreover, the main focus of our paper is on the timing of economic benefits for administrative centers. We show that these gains only materialize in the long run.

Finally, we contribute to a growing literature studying the economic consequences of the Revolution and the Napoleonic period inside and outside of France (Acemoglu et al. 2011; Finley, Franck, and Johnson 2021; Franck and Michalopoulos 2017; Juhász 2018; Postigliola and Rota 2021; Squicciarini and Voigtländer 2016). de Tocqueville (1856) famously argued that the Revolution did not fundamentally disrupt the political dynamics inherited from the Ancien Régime since centralizing trends had already been at work under the monarchy. A large literature across demography, geography, and history has studied the legacy of the creation of French departments (Brette 1907; Ozouf-Marignier 1992; Todd and Le Bras 2012). Our paper is the first in this literature to provide evidence on the consequences of revolutionary reforms at the municipal level. We show that the Revolution set in motion a long-term divergence that benefited newly chosen centers of administrative power, which attracted subsequent investments in state capacity.

The rest of the paper is organized as follows. Section I provides background on the administrative reform and the selection of new capitals. Section II presents our conceptual framework. Section III describes our data and empirical strategy. Section IV presents our results in the short and medium run, while Section V looks at impacts in the long run, in both cases differentiating effects on public goods versus economic development. Section VI concludes.

I. Historical Background

A. *Ancien Régime Divisions*

Before the Revolution, the French Kingdom was governed through many layers of jurisdictions inherited from Roman and medieval institutions. Various types of administrative entities with nonoverlapping boundaries fulfilled state functions at the local level. A complete description of Ancien Régime administrative functions is beyond the scope of this paper, but here we provide a brief summary building on the seminal work of Nordman et al. (1995).

First, France was divided into 136 *évêchés* (bishoprics), which broadly mirrored the boundaries of ancient Roman provinces. Being the seat of a bishopric was a source of considerable prestige. These cities concentrated the local functions of the Catholic Church and remained relevant for other basic functions of the state, such as

censuses (population was counted at the level of parishes). The royal administration was itself divided into entities known as *généralités* and *subdélégations*, competent for various functions, including tax collection and military surveillance. Within the *généralités*, tax collection was organized around local centers known as *recettes des finances*. Fiscal rules differed across two types of jurisdictions known as *pays d'élection*, where the local representative of the Crown enjoyed broad powers, and *pays d'états*, where tax collection was governed by local rules and regulated by an assembly of the three estates. Judicial functions were mostly fulfilled at the level of *bailliages* (bailiwicks), known as *sénéchaussées* in southern France.³ These entities were also used for electoral purposes; the quotas of local representatives at the Estates General of 1614 and 1789 were determined at the level of the bailiwicks.

Overall, France had approximately 800 local administrative centers before the Revolution. Because the attribution of functions across these centers was complex, and because the boundaries between these various entities were often contested, administrative complexity was a major issue in the *cahiers de doléances* (lists of grievances) drafted ahead of the Estates General of 1789. In practice, a single parish could depend on multiple jurisdictions for different services: “the cases of citizens having to visit different towns to face trial and to pay taxes were innumerable” (Nordman et al. 1995).

B. The Creation of Departments

In June 1789, delegates from the Third Estate gathered in Versailles declared themselves the National Assembly and swore an oath to not disband until a new constitution was drafted for the kingdom. This process led to the creation of a constitutional committee (*Comité de constitution*) composed of eight members. The committee played a key role in the design of the new administrative map of the French kingdom between September 1789 and February 1790.⁴

Through a series of decrees culminating in the Decree Relative to the Division of the Kingdom adopted on February 26, 1790, the committee undertook the task of drastically revamping the French administrative map. The original idea of the reform was to break down old provinces and to unify local state functions within a new, pyramidal nomenclature. An early design advocated by Sieyès proposed a partition of the kingdom into a grid with 81 equal-sized cells (online Appendix Figure A.1). In order to ease citizens' access to administrative services, newly established local centers (*chefs-lieux*) would concentrate all the functions of the local state, including fiscal, judicial, administrative, and police functions.

However, breaking up the old administrative order proved a more difficult task than originally envisioned. The process of reallocating administrative functions across cities was fraught with controversies and lobbying attempts by local elites.⁵

³The judicial hierarchy also included higher courts known as *présidial* as well as the *Parlements*, which rendered justice in the name of the king with no possibility of appeal. There were 13 Parlements of unequal size by the end of the Ancien Régime. The largest of these, the Parlement of Paris, had jurisdiction over nearly half of the French territory.

⁴The committee is sometimes referred to as the Sieyès-Thouret committee after its two most influential members: Abbé Sieyès (1748–1836) and Jacques-Guillaume Thouret (1746–1794).

⁵In total, the committee examined more than 10,000 letters and met with 1,884 town emissaries (Nordman et al. 1995).

Ultimately, the committee settled for a compromise solution, which consisted of creating 83 new administrative units of approximately equal size, known as departments (*départements*), some of which followed the boundaries of old provinces, while others did not. This solution preserved two founding principles of the initial proposal, namely the centralization of administrative functions and easy access of all citizens to the local administration, while also appeasing local tensions around the reform. The new departments spanned approximately 18 by 18 leagues (approximately 6,400 km), a size intended to allow citizens to travel to the capital in no more than 1 day.

In practice, the committee pursued two conflicting goals: creating a new homogeneous map of the French territory, while also frequently granting requests from local elites to preserve existing administrative centers. As a result, two types of departments emerged: on the one hand, departments reflecting historical boundaries, usually centered around an old city or province, and on the other hand, more artificial departments created by combining or dividing existing territories, so that the objective of equally sized departments could be attained. Our analysis focuses on the latter departments, as described below.

C. Artificial Departments and the Uncertainty around Capitals

In many of the artificial departments created in January–February 1790, no single city stood out as the obvious candidate to become the new capital. In these instances, the committee received many competing demands from the mayors, delegates, or even citizens of rival cities.⁶ We collected the full set of letters written to request a department capital.⁷ These letters provide a unique perspective on the local rivalries that made it difficult to designate a capital in artificial departments.

Strikingly, the characteristics invoked to claim the status of capital (including population, prior administrative functions, economic potential, and centrality) were not used in a consistent way and were often used to reach opposite conclusions. For example, the criterion of having many prior administrative functions was sometimes viewed positively (no investments would need to be made in cities where administrative buildings already existed) and sometimes presented as a problem, as some letters invoked the necessity to get rid of the old order. A letter in support of the city of Mézières (Ardennes) for instance stated that “the most ill-treated places under the reign of despotism shall be the most favored under that of liberty.”⁸ Claims made about the economic potential of cities were equally ambiguous. In several instances, poorer candidate cities argued that not obtaining the capital status would precipitate their citizens in misery, as in the case of Laon (Aisne): “the just hope of becoming the capital is, for the city of Laon, like a life raft. The city has no trade.”⁹

⁶ Some were even written by representatives from other cities. For example, the mayor of Paris wrote in support of Soissons against Laon in the Aisne department, presenting Soissons as the “granary of the capital.”

⁷ Arch. nat., D IVbis 1-18, Archives du comité de Division du territoire.

⁸ Translated from “*Les endroits les plus maltraités sous le règne du despotisme, doivent être les plus favorisés sous celui de la liberté*” (authors’ translation).

⁹ Translated from “*La juste espérance de devenir chef lieu de département est, pour la ville de Laon, la planche dans le naufrage. Cette ville n’a aucun commerce*” (authors’ translation).

Overall, the letters written in support of the different candidate cities reveal how, in the artificial new departments created as part of the reform, the candidate cities combined advantages and disadvantages. This prompted the Assembly to invent new solutions to defuse local tensions:

Urban rivalries and multiple grievances were delicate to address (...). Resolving these conflicts embarrassed the Committee so much that, seizing upon a proposal by Rabaut Saint-Étienne, several legislative solutions were adopted to circumvent these challenges. On December 9, 1789, the rotation of local assemblies across several cities becomes legally possible; it is also envisioned that different types of public establishments can be based in distinct cities (...). These mechanisms make it possible to hasten the new division [of the kingdom] while silencing competing local claims.

Nordman et al. (1995, vol. 5, p. 54) (authors' translation)

These solutions provide the basis for our identification of the causal impacts of the 1790 reform. Following Rabaut Saint-Étienne's proposed design, the February 1790 decree stipulated that in some departments, administrative functions would rotate across different cities. In a small number of other departments, these functions would be shared across the department's major cities, or the decision was simply adjourned and left to a local vote. A common pattern across these departments was that the members of the Assembly "established no link between the existing urban network and the location of future capitals" (Arnould 1992). Among the 28 departments facing this situation, 17 were provided with an explicit list of candidate cities, while 11 were not (see online Appendix Table A.1).¹⁰

D. The Establishment of Local Capitals

These institutional solutions, in particular the rotation system, rapidly proved impractical and expensive. A new decree abolishing rotations was passed in September 1791, with the bill rapporteur stating,

What is more ridiculous and onerous than these wandering administrations, which require the movement of papers, staff, a proliferation of buildings to accommodate them! Your Constitution has established a direct surveillance over all the points of the Empire, every citizen finds near him an administration that directs him (...). As a result, the exodus of administrations has become not only useless, but also shocking; the time has come to put an end to this abuse.

Preamble to the September 11, 1791 decree (authors' translation)

Article 2 of this decree stipulated that capitals would "remain fixed in the locations in which they now reside." Precise information on the location of administrations in September 1791 is, to the best of our knowledge, unavailable, and we cannot therefore exclude that in some cases, the final choice corresponded to the first city in

¹⁰The February 1790 Decree lists five additional departments in which a rotation was due to take place. We do not include these departments in our analysis since the historical record shows there was no uncertainty with respect to the allocation of the capital, as reflected in the original names of these departments.

the rotation. Our identification strategy requires that the candidate cities were similar based on pre-reform characteristics, thus making the order of the rotation irrelevant. In Tables 2 and 3 (discussed in Section IIID), we provide evidence that final capitals were statistically indistinguishable from other candidate cities along a wide range of baseline characteristics. Online Appendix B provides further background on the choice of candidates and capitals in each department in our sample. The map of French departments and their corresponding capitals has remained largely unchanged for the following two centuries.¹¹

Napoléon gave additional relevance to these new administrative units at the start of the nineteenth century, calling them *préfectures*, appointing civil servants and creating new institutions embodying the local state. These institutions are, for the most part, still in existence today. The prefects sitting in the new capitals were given extensive powers as the sole representative of the state in each department. Napoléon is reported to have said that “the prefects, once they are one hundred leagues away from the capital, have more power than myself” (Reymond 1978, p. 227) and later in his life compared the prefects to “emperors with small feet” (de Las Cases 1824).

Napoléon also rebuilt the local fiscal capacity of the state by creating in each department a fiscal extraction office staffed with a director and tax collectors. In 1807 he launched the project of building municipal cadastres, a task that was only completed in 1850. The cadastre was designed to facilitate the taxation of land property.¹² Finally, Napoléon created France’s modern education system, including high schools (*lycées*) created in May 1802.¹³ To accommodate these new institutions and their employees, a large number of public buildings were built in capital cities. In our analysis, we use the number of public building projects of various types as a measure of the administrative presence of the state.

II. Conceptual Framework

We now present a conceptual framework analyzing how an initial shock in coercive state capacity in otherwise comparable cities affects outcomes in the short and long run. This framework highlights the dynamic and spatial trade-offs faced by a central government in the state-building process. The model is designed to study the evolution of state capacity, public and private goods production, and population.

¹¹The main exceptions to this are the creation of the Tarn-et-Garonne department in 1808, the annexation of Nice and Savoy in 1860, and the boundary changes resulting from the German annexation of Alsace-Lorraine between 1870 and 1918. In addition, the capital of the Charente-Inférieure department was moved from Saintes to La Rochelle in 1810, and the capital of Var was moved from Draguignan to Toulon in 1974 and had moved around candidate cities between 1790 and 1800. Our results are robust to dropping these two departments from the sample.

¹²The idea to create a national cadastre emerged in 1791 to help the government to levy a new “property tax distributed by proportional equality on all built and non-built properties” (Clergeot 2007). From August 1791, communes were authorized to carry out land-surveying operations to improve the “knowledge of the economic resources of each administrative unit.” These operations were financed by the central government until 1821 and by departments and municipalities afterward.

¹³Other institutions relevant at the national level, but less so for the local one, were also created at the time. These include, among others, the Central Bank in 1800 and the Civil Code in 1804.

A. Model

Setting.—Consider two cities $\{c, r\}$, where c is the chosen capital city and r an initially comparable city. There are two periods $t \in \{1, 2\}$. In period t , city j is characterized by its level of local public goods $G_{j,t}$ available only in j , private goods $Q_{j,t}$, coercive capacity $C_{j,t}$, tax revenue $T_{j,t}$, population $N_{j,t}$, and wages $w_{j,t}$. Global public goods $\bar{G}_t$ are available in both cities. c and r are perfectly comparable at the beginning of period 1, except that c receives a positive shock in coercive capacity, as described below.

Public and Private Goods.—Private goods are produced according to the linear production function $Q_{j,t} = A_{j,t}N_{j,t}$, where the productivity in city j and period t is given by $A_{j,t} = A(G_{j,t})^{\beta_1}$. This linear production function assumes that local public goods (schools, infrastructure, hospitals) increase the productivity of the city (as in Guillouzouic, Henry, and Monras 2021). The parameter β_1 captures the intensity of this effect. We assume that the private sector is competitive so that workers are paid their marginal product of labor: $w_{j,t} = A_{j,t}$.

Public goods are produced using tax revenues. We denote as $\mu_{j,t}$ the share of revenue allocated to local public goods, and $\nu_{j,t}$ the share of revenue allocated to global public goods, where $\mu_{j,t}$ and $\nu_{j,t}$ will be choice variables. The rest of the tax revenue $(1 - \mu_{j,t} - \nu_{j,t})$ is spent on building coercive capacity.¹⁴

Coercive Capacity and Taxes.—Raising taxes requires coercive capacity, which we conceptualize as a stock variable $C_{j,t}$ defined at the local level. An increase in coercive capacity allows the government to raise taxes more efficiently; the final tax revenue available to the state is given by $T_{j,t} = C_{j,t}\tilde{T}_{j,t}$, where $\tilde{T}_{j,t} = \tau_{j,t}N_{j,t}w_{j,t}$ are the resources extracted in j , $\tau_{j,t}$ standing for the individual tax rate. Thus, in the absence of coercive capacity ($C_{j,t} = 0$), tax revenue cannot be collected.

Building coercive capacity requires investments. We assume the following production function for coercive capacity: $C_{j,t} = C_{j,t-1}(B_{j,t-1})^\sigma$; that is, coercive capacity increases proportionally to the budget amount allocated to it, where $B_{j,t} = (1 - \mu_{j,t} - \nu_{j,t})T_{j,t}$. Before the beginning of the first period, the capital city receives a shock C in coercive capacity, so that $C_{c,1} = C_{r,1} + C$.

Preferences.—There are two types of actors in this environment: the central government and a mass of citizens N . There is no overall population growth. Citizen i in city j maximizes her utility given by

$$U_j^t(i) = (Q_j^t)^{\gamma_1} (G_j^t)^{\beta_2} (\bar{G}^t)^{\beta_3} e^{\varepsilon_j^t(i)},$$

under the budget constraint $(1 - \tau_j^t)w_j^t = Q_j^t$ (the private good is the numeraire). ε_{ijt} is an individual taste shock drawn from a type I extreme value distribution. The

¹⁴ Local public goods are produced using local tax revenues according to $G_{j,t} = \mu_{j,t}T_{j,t}$. Global public goods are produced combining resources from both cities, $\bar{G}_t = (R_{c,t})^{1/2}(R_{r,t})^{1/2}$, where $R_{j,t} = (\nu_{j,t}T_{j,t})^{\alpha_j}$. This captures the fact that collecting resources from each city requires coercion and is therefore costly. This is the second difference between the capital and the candidate city: we assume $\alpha_c > \alpha_r$.

citizens are fully mobile every period and choose the city that maximizes their log indirect utility. However, we introduce an element of limited rationality: citizens who move in period t decide to move based on current conditions, even though they will face different conditions (taxes, public goods, etc.) in the next period.¹⁵ We denote V_j^t the component of log indirect utility independent of the taste shock.¹⁶

Finally, we assume that the government maximizes the discounted sum of weighted log indirect utilities of citizens.¹⁷ The discount factor is given by δ , and the government sets weight ξ on the welfare of the capital. Thus, the government chooses tax rates and the allocation of tax revenues to maximize

$$\xi V_{c,1} + (1 - \xi) V_{r,1} + \delta [\xi V_{c,2} + (1 - \xi) V_{r,2}].$$

Timing.—In each period the timing is the following: (i) the government announces taxes as well as the allocation of tax revenue; (ii) taxes are raised, coercive capacity is realized, and public and private goods are produced; and (iii) citizens draw their individual taste shock and make their location choice.

B. Predictions

PROPOSITION 1: *In equilibrium,*

- (i) *In Period 1, (a) taxes are strictly higher in the capital; (b) if the initial shock in coercive capacity for the capital is not too large, $C < \bar{C}$, the capital has fewer local public goods, the private sector is less productive and pays lower wages; and (c) population migrates away from the capital.*
- (ii) *In Period 2, (a) the gap in coercive capacity has grown ($\ln C_{c,2} - \ln C_{r,2} > \ln C_{c,1} - \ln C_{r,1}$); (b) taxes are higher in the capital; and (c) if $\beta_3 \leq \bar{\beta}_3$, more local public goods are provided in the capital, the private sector is more productive and pays higher wages, and population migrates to the capital.*

Coercive capacity is the key ingredient in the model. Proposition 1 shows that the initial shock in coercive capacity in favor of c is amplified in period 2. In period 1, the government chooses the size and the allocation of the budget in both cities. We show in the Proof of Proposition 1 (see online Appendix) that the budget share allocated to coercive capacity is the same in c and r . Intuitively, the decision of allocating funds to coercive capacity amounts to a question of budget allocation across periods and is the same in both cities. This decision depends on the government's discount factor δ and on how fast coercive capacity grows with investments σ , but not on the taste of citizens for public and private goods.

The budget allocation between coercive capacity and public goods is the same in both cities, but the size of the budget differs for two reasons. First, because it

¹⁵This does not change the results but simplifies the resolution since the central government in setting taxes does not need to take into account the effect it might have on population in the current period.

¹⁶The log indirect utility can be written $\gamma_1 \ln(Q_j^t) + \beta_2 \ln(G_j^t) + \beta_3 \ln(\bar{G}^t) + \varepsilon_j^t(i) \equiv V_j^t + \varepsilon_j^t(i)$.

¹⁷We show in the online Appendix that the model can accommodate a government seeking to extract rents.

is easier for the government to recover taxes in the capital, the government sets a higher tax rate in c . Second, the disposable budget is higher in c since, for the same level of taxes, more resources are gathered due to the initial shock C . These choices imply that the gap in coercive capacity grows between the two cities between periods 1 and 2.

These dynamics in the buildup of coercive capacity determine the shape of the equilibrium described in Proposition 1. Initially, the capital raises more taxes to build coercive capacity and to fund global public goods. This drives citizens away from the capital, unless the initial advantage in coercive capacity is so large that it allows the capital to also fund more local public goods. The gap in coercive capacity then grows larger than the initial shock. In period 2, the capital raises more tax revenue due to its increased coercive capacity. These resources fund local public goods as well as global public goods. In spite of higher taxes, population moves to the capital, attracted by local public goods and the higher productivity of firms and thus wages, provided not too much of the collected funds are allocated to global public goods (i.e., a condition on parameter β_3).

Overall, this framework shows how dynamics in the public and private sectors as well as population in the capital, relative to candidate cities, are driven by the dynamic buildup of coercive capacity. We use these predictions to guide our empirical explorations. We, however, highlight that we do not view the empirical results as a direct test of the model, and we cannot exclude alternative channels for some empirical results.¹⁸

III. Empirical Framework

Section II predicts different dynamics for the capital and other candidate cities once the state-building process is set in motion. After receiving more coercive capacity initially, the capital eventually funds more public goods and may as a consequence experience stronger economic growth and host a larger population. However, this divergence only materializes in the long run. We now describe the empirical strategy we use to study these dynamic effects of the state-building project embodied in the 1790 reform.

A. Sample Construction

Our analysis focuses on the artificial departments that were not deliberately constructed around a single city or province, and where the Assembly instructed that the capital should be chosen among multiple candidate cities, as described in online Appendix Table A.1. Starting with the complete list of 83 departments, we first systematically identify, based on their original names, the 43 departments that were explicitly constructed around a single major city, a royal province, or the subset of

¹⁸For instance, while our model predicts that productivity gains in the capital should materialize only in the medium term as a response to the provision of higher-quality public goods, it is challenging to distinguish this in the data from a technological shock in period 2. We, however, provide evidence in Section IVE showing that technological shocks are unlikely to drive our main results.

a royal province that included the old provincial capital.¹⁹ These departments are excluded from our analysis.

Among the remaining artificial departments, we then identify 28 departments for which the Assembly failed to explicitly designate a new capital and resorted to using rotations or local elections to expedite the reform.²⁰ Among these, 17 departments were provided with an explicit list of candidate cities, while 11 were not. Our main sample, later on referred to as baseline sample, includes the former 17 departments, and we consider as candidate cities all the 50 cities specified in the official decrees passed in January–February 1790. We then reconstruct the list of likely candidate cities for 8 additional departments using information contained in additional decrees passed after February 1790, petitions addressed to the Assembly, votes held in local assemblies, and the historical record in Masson (1984) and Margadant (1992), as described in online Appendix Table A.2.²¹ We do not include the three remaining departments since there is no evidence of competition for the capital in these departments. We refer to the sample of 69 cities across 25 departments as our augmented sample. Table 1 provides the complete list of candidate cities across both samples, and Figure 1 shows the spatial distribution of the 17 and 25 departments in our baseline and augmented samples, respectively.

Within all the departments we consider, in order to estimate effects on the periphery of candidate cities, we collected data to cover the universe of municipalities. The list of municipalities comes from the eighteenth-century topographic map of the French kingdom, known as the Cassini map (Pelletier 1990, 2002). A research team from EHESS digitized complete data on Cassini villages mapped to contemporary French municipalities. These data include detailed information on past and present town names, administrative identifiers, land area, altitude, latitude and longitude, mergers and splits, and demographic data since 1793 (Cassini and EHESS 2021).²² Using these data, we construct a balanced panel of municipalities covering 1800 to 1914. Most municipalities experience no boundary change during this period. For the remaining municipalities, we reconstruct municipal clusters combining towns and villages that either merged or separated from each other between 1800 and 1914.²³ The 25 departments of interest include 10,892 municipalities (out of 36,360 French municipalities), including 1,088 municipal clusters (10 percent).

¹⁹ Given the maze of administrative functions described above, defining France's provinces as of 1789 is not straightforward (see Brette 1907). Following Masson (1984), we use the list of 39 provinces provided in the Royal Ordinance of March 8, 1776.

²⁰ One additional department, Rhône-et-Loire, was dissolved as early as 1793 and is therefore excluded from our analysis.

²¹ A local election took place in five of our departments of interest, and all but one of these votes appear to have been contentious. In Drôme, the city of Valence (157 votes) defeated Montélimar (140) and Crest (68) in a three-way vote. In Manche, the vote was evenly split between Coutances and Saint-Lô. In Seine-et-Marne, Melun was chosen over Meaux in a 259–231 vote. In Aisne, Laon defeated Soissons in a landslide (411 to 37) after the Soissons delegates boycotted the election. In Indre, Châteauroux defeated Issoudun by 262 to 47 votes.

²² More information on the Cassini database is available at http://cassini.ehess.fr/fr/html/5_donnees.htm.

²³ Clusters are named after the absorbing commune (in cases of mergers) or the parent commune (in cases of splits). When a new commune was carved out of two or more communes, the cluster is composed of the child and all the parent communes.

TABLE 1—SAMPLE DEPARTMENTS AND CANDIDATE CITIES

Department	Candidate cities
Aisne	Laon ^a , Soissons
Ardèche	Annonay, Aubenas, Bourg-Saint-Andéol, Privas ^a , Tournon
Ariège	Foix ^a , Pamiers, Saint-Girons
Cantal	Aurillac ^a , Saint-Flour
Charente-Inférieure	Saint-Jean-d'Angély, La Rochelle, Saintes ^a
Deux-Sèvres	Niort ^a , Parthenay, Saint-Maixent
Dordogne	Bergerac, Périgueux ^a , Sarlat
Haute-Marne	Chaumont ^a , Langres
Haute-Saône	Gray, Vesoul ^a
Indre	Châteauroux ^a , Issoudun
Jura	Dole, Lons-le-Saunier ^a , Poligny, Salins-les-Bains
Lozère	Marvejols, Mende ^a
Meuse	Bar-le-Duc ^a , Saint-Mihiel
Saône-et-Loire	Chalon-sur-Saône, Mâcon ^a
Tarn	Albi ^a , Castres
Var	All district capitals (9 cities)
Vosges	Épinal ^a , Mirecourt
Ardennes	Charleville, Mézières ^a , Rethel, Sedan
Aveyron	Rodez ^a , Villefranche-de-Rouergue
Drôme	Crest, Montélimar, Valence ^a
Finistère	Landerneau, Quimper ^a
Landes	Dax, Mont-de-Marsan ^a
Manche	Coutances, Saint-Lô ^a
Oise	Beauvais ^a , Compiègne
Seine-et-Marne	Meaux, Melun ^a

Notes: This table lists all the artificially created departments for which the capital status remained uncertain as of February 1790, along with the corresponding candidate cities. In our baseline analysis, we only consider the 17 departments for which the Constituent Assembly specified the list of candidate cities. The eight additional departments in the bottom panel are those for which we reconstruct the list of candidate cities using information contained in additional decrees passed after February 1790, petitions addressed to the Assembly, and the historical record in Masson (1984) and Margadant (1992). Online Appendix Table A.2 provides additional details. Capitals chosen as of 1800 are flagged with an “a”. Online Appendix B provides detailed historical background on the choice of candidate cities and capitals in each department.

B. Data

We draw upon several data sources to explore the evolution of state capacity in the short and long run. A key novelty of our approach is to assemble data at the most granular administrative level, namely the municipality, along a large number of dimensions.

Old Administrative Functions and Police Force.—Nordman et al. (1995, pp. 74–80) provide city-level information on the location of the four main administrative functions under the Ancien Régime: the *évêchés* (bishoprics), the *bailliages* (bailiwicks), the *recettes des finances* (tax centers), and the *subdélégations*

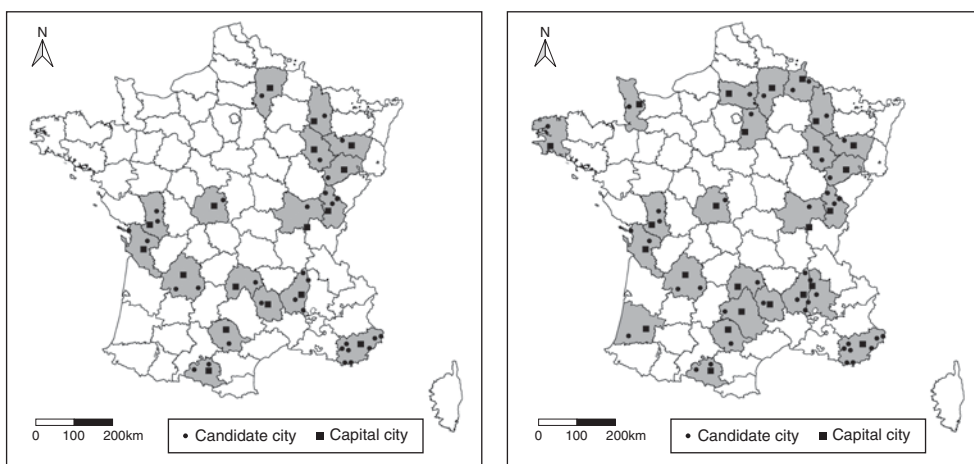


FIGURE 1. ARTIFICIAL DEPARTMENTS WITH MULTIPLE CANDIDATE CAPITALS IN 1790

Notes: This map shows the departments in our baseline sample (17 departments, left-hand side) and augmented sample (25 departments, right-hand side), colored in gray. Black lines indicate the boundaries of French departments created in February 1790 (see Chambru 2020). Capital cities after 1800 are indicated with a square. Other candidate cities are indicated with a circle.

(administrative centers). By 1789, there were approximately 136 bishoprics, 426 bailiwicks, 339 tax centers, and 686 administrative centers across the French territory. We rely on these variables for balance checks, and we show that our results are robust to controlling for the pre-reform presence of administrative functions.

In addition, we separately collected data on the location of police units (known as *maréchaussée*) and the location of secondary schools in 1789–1790. Originally established to oversee royal troops, the *maréchaussée* was abolished in the early stages of the Revolution and reinstated in 1791 under the name *Gendarmerie Nationale*. We rely on the *Atlas Historique de la Gendarmerie* to collect information on the location of every brigade in 1790 (Fressin 2022). We further used the list provided in the *Almanach royal* to gather information on the location of gendarmerie brigades in 1816.²⁴

Coercive Capacity.—To measure investments in local coercive capacity, we collected data on the establishment of the cadastre (*cadastre napoléonien*), military conscription between 1802 and 1815, business taxes collected in 1839, and the construction of tribunal and prison buildings until 1840.

First, we used the inventories from departmental archives (series P and 3 P) to collect information on the year of the first cadastre established in each municipality. Overall, we retrieved cadastral data for 97 percent of municipalities in our sample. Since the date of the original cadastre is additionally missing for 5 percent of these municipalities, we do not observe the date of the first cadastre for 607 municipalities (8 percent) in our baseline sample. As discussed in Section ID, the establishment

²⁴ For the *Atlas Historique*, see <https://atlas-gendarmerie.fr/>. For the *Almanach*, see Bibliothèque Nationale de France (2021, pp. 546–56).

of municipal cadastres took place over a 50-year period starting in 1807; by 1850, 99 percent of municipalities had their cadastre established.

To measure impacts on military enrollment, we rely on data from the collaborative online project *Matricules Napoléoniens* (2021). This project digitized enrollment records for conscripts who served in Napoleonic armies between 1802 and 1815. The data were collected on a regiment-by-regiment basis and included more than 950,000 soldiers as of 2021. We use information on departments and villages of birth, and we aggregate these data to the municipality level.

For prisons and tribunals, we rely on data from the *Conseil des Bâtiments Civils* created in 1795.²⁵ The council examined all construction, renovation, and maintenance projects of public buildings above a relatively low cost threshold.²⁶ We collect the number of approved projects, focusing in particular on prisons and tribunals, and aggregate these data at the municipality-year level. Finally, for the business tax collected from firms, we rely on data from Chanut et al. (2000), as described below.

Public Goods.—We combine various sources to measure the state's ability to deliver public goods in the aftermath of the 1790 reform. First, we collected data on secondary schools at the municipal level in 1812 and 1836.²⁷ The data include information on locally funded schools (*collèges municipaux*) and centrally funded schools (*lycées*). Section IV provides additional background on these establishments. We additionally use the *Conbavil* data described above to capture investments in hospitals. Second, we digitized data from Bucquet (1874) on local welfare offices (*bureaux de bienfaisance*) in 1871. The welfare offices were established by a 1796 law and provided relief and health care to the sick and the elderly. Services were funded through donations and organized by each city on a voluntary basis. Bucquet's (1874) survey includes approximately 13,000 offices across France (see also Haudebourg 1998, pp. 145–88). Third, we created a historical GIS for French communication and transportation networks around the mid-nineteenth century. We geocoded the 1863 map of the telegraph network from de Vougy (1863) and use interdepartmental connections to measure access to the national telegraph network. We further digitized various historical maps to geocode all French railway stations in 1852 and 1870 (Andriveau-Goujon 1852; Chaix 1870), which we link with our dataset of French municipalities.

Financial and Industrial Development.—Our data on private sector activity include information on industrial activity in 1839–1847, banks, and patents registered throughout the nineteenth century. First, we leverage information from the country's first industrial census conducted between 1839 and 1847. This census provides establishment-level data for 14,238 plants across the territory. We use the microdata from Chanut et al. (2000) and additionally digitized information on the location of establishments at the municipality level, which is available for all plants except those spanning multiple cities (Ministère de l'Agriculture et du Commerce

²⁵ For the *Conbavil* database, see <https://www.inha.fr/fr/ressources/outils-documentaires/conseil-des-batiments-civils-conbavil.html>.

²⁶ The threshold for consideration by the council was 5,000 francs after 1806 and 20,000 francs after 1821. This threshold approximately corresponds to the cost of a small classroom.

²⁷ Arch. nat., F 17 6835, Enseignement secondaire, Statistique, 1837–1854.

1847).²⁸ Second, we derived information on bank locations from Hoffman, Postel-Vinay, and Rosenthal (2019).²⁹ We only consider private banks, thus excluding local branches of the central bank (*Banque de France*) as well as *comptoirs d'escompte*, another type of public bank. For patents, we rely on data from the National Institute of Industrial Property, which contains information on the registration date and the municipality of residence of each patent applicant (INPI 2014).³⁰ We compute the aggregate number of patents registered by municipality and by year throughout our study period.

Contemporary Outcomes.—We use contemporary measures of public and private employment to evaluate the reform's economic impacts in the very long run. This information is collected from the *Déclarations Annuelles des Données Sociales* (DADS), matched employer-employee data provided by the French statistics agency (INSEE 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017). These data are built from mandatory social security contributions reported by firms based in France over the period 2006–2015. Public sector employment falls into three categories: national public servants (*Fonction Publique de l'état*, or FPE), local civil servants (*Fonction Publique Territoriale*, or FPT), and hospital workers (*Fonction Publique Hospitalière*, or FPH). These three categories represent approximately 50 percent, 30 percent, and 20 percent of the public sector workforce, respectively.

C. Empirical Strategy

Our empirical strategy consists in comparing, in artificial departments created in January–February 1790, capitals after 1800 with other candidate cities. Under the assumption that the choice of the final capital among the candidate cities was plausibly exogenous to preexisting city characteristics and trends, this strategy allows us to estimate the causal impact of administrative presence on the capitals themselves.

The setting is analogous to a stratified randomization design where the strata are departments and the units of randomization are candidate cities. Within each department, we assume that each candidate city had the same probability of becoming the capital, equal to $1/n_j$, where n_j is the number of candidate cities in department j . Our exploration thus begins with the simple treatment-control comparison:

$$(1) \quad y_{ij} = \alpha_0 + \alpha_1 C_{ij} + \delta_j + \varepsilon_{ij},$$

where i denotes a candidate city, C_{ij} denotes capital status from 1800 onward, and δ_j are department fixed effects. We initially use only the candidate cities to estimate this comparison, using either our baseline sample of $N = 50$ cities in 17 departments or our augmented sample of $N = 69$ cities in 25 departments. We report estimates

²⁸ A second industrial census was conducted in 1860–1865. Chanut et al. (2000) provide the microdata for this survey. Unfortunately, we could not locate the corresponding information on the municipality of operation for each industrial establishment.

²⁹ See https://didomena.ehess.fr/concern/data_sets/zp38wd010?. The data contain information on the opening of local bank branches at the city level at regular time intervals. We exclude the year 1864 from the raw data from Hoffman, Postel-Vinay, and Rosenthal (2019) because of data quality issues.

³⁰ See <https://www.inpi.fr/fr/open-data-brevets-19eme>.

obtained from two different approaches. The first approach follows a design-based perspective: we aggregate treatment and counterfactual outcomes (measured across nonchosen candidate cities) within each department, and we perform a paired t -test with $K - 1$ degrees of freedom, where $K \in \{17, 25\}$ is the number of departments, treating the setting as a stratified randomization design. We then conduct randomization inference based on this t -statistic. Our baseline estimation uses the unweighed data at the city level; in the online Appendix, we also show results obtained after reweighing the data with the inverse of the conditional probability of becoming the capital inside department j ($p_{ij} = 1/n_j$) or the inverse probability that a city i falls within stratum j ($p_j = n_j/N$).

Second, we estimate (1) parametrically via OLS, and we construct 95 percent permutation-based randomization confidence intervals around α_1 , using the robust approach of DiCiccio and Romano (2017) with 50,000 permutations of the treatment (see also Young 2024). This approach allows us to additionally include controls, which we do in online Appendix results. In our baseline estimation, the standard errors from equation (1) are heteroskedasticity robust; in the online Appendix we show robustness to using standard errors clustered by department.

We use a similar approach to estimate the impact of proximity to the final capital among neighboring municipalities. In this case, the estimating equation is

$$(2) \quad y_{ij} = \beta_0 + \beta_1 \text{CloseTo } C_{ij} + X_{ij}\Omega' + \delta_j + \varepsilon_{ij},$$

where $\text{CloseTo } C_{ij}$ is a binary variable equal to 1 if a municipality's nearest candidate city is chosen as capital, and the sample contains all the municipalities located within a k -km radius around candidate cities, excluding the candidate cities themselves. We additionally include a set of pre-reform geographic, demographic, and administrative controls X_{ij} .³¹ We use $k = 20$ km in our baseline estimation and show robustness to alternative values of k in the online Appendix. Standard errors are clustered at the department level. Using this approach, we again report 95 percent permutation-based randomization confidence intervals obtained using 50,000 permutations of the treatment. In conducting these permutations, we define the treatment at the level of candidate city i , analogous to a clustered randomization where the clusters are blocks containing all municipalities in the periphery of city i .

Finally, we jointly estimate the impact of capital status and proximity to the capital:

$$(3) \quad y_{ij} = \gamma_0 + \gamma_1 C_{ij} + \gamma_2 \text{CloseTo } C_{ij} + \gamma_3 R_{ij} + X_{ij}\Omega' + \delta_j + \varepsilon_{ij},$$

where C_{ij} and $\text{CloseTo } C_{ij}$ are defined as before. R_{ij} is a dummy equal to 1 for all candidate cities and captures the endogeneity associated with potential capital status and its correlates in terms of administrative and market potential. Here, the sample includes all municipalities located within a 20 km radius of candidate cities as well

³¹The geographic controls used in all our specifications include land area and land area squared, minimal and maximal altitude, latitude and longitude. In Table 8, we additionally control for log population in 1793 and 1800 and four dummy variables indicating the presence of each of the four main administrative functions under the Ancien Régime (bishoprics, bailiwicks, tax centers, and *subdélégations*), distance to the department centroid, pre-reform market access, and wheat suitability.

as the candidates themselves. The parameters of interest that are causally identified (conditional on R_{ij}) in this equation are γ_1 and γ_2 . In this approach, we report (i) standard errors clustered by candidate city, which is the level at which the treatments C_{ij} and $CloseTo C_{ij}$ are defined; (ii) standard errors clustered by department, in which case we also report the p -values from a wild bootstrap to account for the small number of departments (Cameron, Gelbach, and Miller 2008).

D. Identification Checks

Section IB discusses how, in a subset of artificially created departments, the Assembly faced difficult choices among rival candidate cities and ultimately failed to designate a single city as capital. In some of these departments, a rotation of administrative functions between candidate cities was established; in others, this choice was postponed and left to a local vote. These design choices were made to hasten the adoption of the reform under substantial pressure from local representatives.

A comparison between the artificial departments in our sample and other departments, presented in online Appendix Table A.3, supports this interpretation. The artificial departments in our baseline and augmented samples are slightly larger and less densely populated and are similar in terms of the presence of Ancien Régime administrative functions. Three key differences, however, stand out. First, the largest city in artificial departments is, on average, four times smaller than the largest city in out-of-sample departments (three times larger if we exclude Paris). Second, the size ratio between the first- and second-largest city is approximately ten times larger in nonartificial departments (six times larger if we exclude Paris). This is consistent with the historical evidence, described in online Appendix B, that no obvious single candidate city stood out in artificial departments, relative to others. Third, the chosen capital is much more centrally located in nonartificial departments, compared to artificial departments.

Our identification strategy is further validated by a series of balance checks. In Tables 2 and 3, we compare capitals and other candidate cities along a range of pre-reform characteristics in our baseline and augmented samples, respectively. These variables include two baseline measures of market access via roads and navigable waterways defined as in Donaldson and Hornbeck (2016), and assuming either low or high trade frictions,³² the related indicator of urban potential from Bosker, Buringh, and van Zanden (2013), which is the sum of inverse distance-weighted municipal populations computed using the universe of municipalities in 1793;³³ log

³² Following Donaldson and Hornbeck (2016), we construct a network database of roads and waterways before the French Revolution and calculate lowest-cost routes between each municipality in our sample and towns larger than 1,000 inhabitants. We extract data on the road network in 1789 from Arnaud (2021). We derive information on navigable rivers and canals from Dupain-Triel (1781), which we complete with data on active trade ports in France from the Portic Project (<https://anr.portic.fr/en>). To assign the costs of transportation for each network, we rely on figures used by the Revolutionary government in 1790 (Daudin 2007, p. 8). We further assume that moving goods outside the road network was twice as costly as moving goods on roads. Finally, we assume either low trade elasticity ($\theta = 3$ as in Donaldson and Hornbeck 2013) or high trade elasticity ($\theta = 8.22$ as in Donaldson and Hornbeck 2016).

³³ We define urban potential for city i as the distance-weighted sum of the size of all other French municipalities: $UP_i = \sum_j^n \frac{pop_j}{\max\{D_{ij}, 1\}}$, where pop_j is the population of city j in 1793 and D_{ij} is the distance between city i and city j . Information on 1793 population is missing for 853 municipalities (2.3 percent). We substitute it with information on population in 1800.

TABLE 2—BALANCE BETWEEN CAPITALS AND OTHER CANDIDATES: BASELINE SAMPLE

	Capitals [SD] (1)	Candidates [SD] (2)	<i>t</i> -test {RI <i>p</i> -value} (3)	Difference (SE) (4)	OLS (SE) (5)	Rand. inference [95% CI] and { <i>p</i> -value} (6)
<i>Panel A. Demography</i>						
Market access (low trade frictions)	10.46 [0.55]	10.47 [0.66]	<i>t</i> = −0.17 {0.867}	−0.016 (0.094)	−0.041 (0.098)	[−0.22, 0.15] {0.82}
Market access (high trade frictions)	6.50 [1.55]	6.31 [1.74]	<i>t</i> = 0.52 {0.608}	0.187 (0.359)	0.139 (0.374)	[−0.64, 0.80] {0.84}
log urban potential	11.66 [0.13]	11.66 [0.13]	<i>t</i> = 0.38 {0.709}	0.004 (0.010)	0.004 (0.010)	[−0.02, 0.19] {0.41}
log population in 1793	8.85 [0.44]	8.75 [0.49]	<i>t</i> = 0.86 {0.409}	0.101 (0.117)	0.095 (0.120)	[−0.13, 0.51] {0.23}
log population in 1800	8.86 [0.41]	8.77 [0.42]	<i>t</i> = 0.80 {0.460}	0.087 (0.109)	0.088 (0.116)	[−0.13, 0.49] {0.25}
Population growth, 1793–1800	0.01 [0.17]	0.02 [0.16]	<i>t</i> = −0.27 {0.792}	−0.013 (0.050)	−0.007 (0.041)	[−0.09, 0.07] {0.72}
Largest city in department (0/1)	0.35 [0.49]	0.34 [0.42]	<i>t</i> = 0.09 {0.935}	0.017 (0.198)	−0.006 (0.162)	[−0.32, 0.44] {0.74}
Land area (km ²)	33.40 [19.04]	37.70 [31.03]	<i>t</i> = −0.46 {0.653}	−4.295 (9.310)	−5.304 (8.331)	[−29.17, 10.61] {0.32}
<i>Panel B. Administrative functions</i>						
Bishopric (<i>évêché</i>)	0.35 [0.49]	0.40 [0.44]	<i>t</i> = −0.43 {0.815}	−0.051 (0.119)	−0.049 (0.111)	[−0.26, 0.27] {0.77}
Bailiwick (<i>bailliage</i>)	0.82 [0.39]	0.84 [0.32]	<i>t</i> = −0.13 {1.00}	−0.015 (0.114)	−0.009 (0.112)	[−0.25, 0.39] {0.48}
Tax center (<i>recette</i>)	0.94 [0.24]	0.75 [0.40]	<i>t</i> = 2.26 {0.060}	0.191 (0.084)	0.203 (0.085)	[0.00, 0.47] {0.00}
Admin center (<i>subdélégation</i>)	0.94 [0.24]	0.90 [0.26]	<i>t</i> = 0.54 {0.745}	0.037 (0.068)	0.021 (0.067)	[−0.16, 0.23] {0.66}
Old functions	3.06 [0.97]	2.90 [1.14]	<i>t</i> = 0.57 {0.729}	0.162 (0.284)	0.166 (0.246)	[−0.35, 1.15] {0.29}
Police force (<i>maréchaussée</i>)	3.76 [0.97]	4.06 [1.95]	<i>t</i> = −0.56 {0.648}	−0.294 (0.527)	−0.207 (0.513)	[−1.00, 1.27] {0.63}
College (<i>collège</i>)	0.94 [0.24]	0.89 [0.27]	<i>t</i> = 0.55 {0.748}	0.051 (0.093)	0.036 (0.108)	[−0.16, 0.29] {0.53}
<i>Panel C. Economic potential</i>						
Wheat suitability (GAEZ)	6.12 [1.12]	6.44 [1.15]	<i>t</i> = −1.42 {0.181}	−0.318 (0.224)	−0.257 (0.234)	[−0.69, 0.67] {0.96}
Caloric yield	1.94 [0.30]	1.94 [0.32]	<i>t</i> = 0.23 {0.828}	0.007 (0.029)	0.009 (0.029)	[−0.22, 0.06] {0.57}
Salt office (<i>bureau des traites</i>)	0.29 [0.47]	0.21 [0.35]	<i>t</i> = 0.70 {0.535}	0.081 (0.115)	0.046 (0.116)	[−0.19, 0.29] {0.56}
Salt granary (<i>grenier à sel</i>)	0.18 [0.39]	0.23 [0.40]	<i>t</i> = −0.55 {0.748}	−0.051 (0.093)	−0.073 (0.084)	[−0.24, 0.12] {0.47}
Height of conscripts	170.35 [1.94]	169.99 [1.33]	<i>t</i> = 0.01 {0.991}	0.005 (0.448)	0.169 (0.719)	[−1.60, 1.56] {0.84}

Notes: This table reports baseline characteristics for our baseline sample of 17 departments measured in capital cities (column 1) compared to other candidate cities (column 2). Standard deviations are reported in brackets. Column 3 reports the *t*-statistic from a paired *t*-test of equality of means between capitals and other candidates, with $N - 1$ degrees of freedom, where $N = 17$ is the number of departments. In curly brackets, we report the randomization inference (RI) *p*-value based on this *t*-stat obtained after 10,000 permutations of the treatment. Column 4 reports the average difference between capitals and other candidates. Column 5 reports the coefficient from regressions of each variable on capital status, where each regression includes department fixed effects (equation (1)). Heteroskedasticity-robust standard errors are reported in parentheses. The *F*-stat from the test of joint significance of all coefficients in column 5 is $F = 1.02$ ($p = 0.488$). Column 6 reports 95 percent permutation-based randomization confidence intervals and (in curly brackets) the randomization inference *p*-value testing the significance of the studentized treatment effect reported in column 5, obtained after 50,000 permutations of the treatment. Across all columns, the sample includes 50 cities in 17 departments.

TABLE 3—BALANCE BETWEEN CAPITALS AND OTHER CANDIDATES: AUGMENTED SAMPLE

	Capitals [SD] (1)	Candidates [SD] (2)	<i>t</i> -test {RI <i>p</i> -value} (3)	Difference (SE) (4)	OLS (SE) (5)	Rand. Inference [95% CI] and { <i>p</i> -value} (6)
<i>Panel A. Demography</i>						
Market access (low trade frictions)	10.62 [0.65]	10.71 [0.71]	<i>t</i> = −1.08 {0.301}	−0.086 (0.080)	−0.090 (0.078)	[−0.23, 0.19] {0.96}
Market access (high trade frictions)	6.88 [1.57]	6.95 [1.73]	<i>t</i> = −0.24 {0.815}	−0.070 (0.287)	−0.048 (0.295)	[−0.59, 0.61] {0.91}
log urban potential	11.66 [0.14]	11.65 [0.15]	<i>t</i> = 1.27 {0.217}	0.013 (0.010)	0.012 (0.009)	[−0.01, 0.15] {0.15}
log population in 1793	8.85 [0.43]	8.76 [0.44]	<i>t</i> = 0.83 {0.419}	0.085 (0.102)	0.077 (0.106)	[−0.11, 0.41] {0.24}
log population in 1800	8.84 [0.43]	8.78 [0.40]	<i>t</i> = 0.66 {0.525}	0.066 (0.099)	0.063 (0.106)	[−0.12, 0.39] {0.29}
Population growth, 1793–1800	−0.01 [0.16]	0.01 [0.14]	<i>t</i> = −0.52 {0.610}	−0.019 (0.037)	−0.014 (0.032)	[−0.09, 0.05] {0.49}
Largest city in department (0/1)	0.32 [0.48]	0.28 [0.40]	<i>t</i> = 0.25 {0.803}	0.038 (0.151)	0.024 (0.132)	[−0.22, 0.34] {0.51}
Land area (km ²)	36.43 [31.09]	36.59 [30.46]	<i>t</i> = −0.02 {0.987}	−0.159 (9.227)	−0.423 (8.486)	[−22.61, 16.12] {0.64}
<i>Panel B. Administrative functions</i>						
Bishopric (<i>évêché</i>)	0.40 [0.50]	0.40 [0.46]	<i>t</i> = 0.04 {1.00}	0.005 (0.135)	0.011 (0.117)	[−0.20, 0.34] {0.67}
Bailiwick (<i>bailliage</i>)	0.84 [0.37]	0.76 [0.37]	<i>t</i> = 0.78 {0.478}	0.077 (0.098)	0.068 (0.097)	[−0.12, 0.40] {0.37}
Tax center (<i>recette</i>)	0.92 [0.28]	0.72 [0.40]	<i>t</i> = 2.42 {0.024}	0.203 (0.084)	0.199 (0.088)	[0.07, 0.44] {0.02}
Admin center (<i>subdélégation</i>)	0.96 [0.20]	0.88 [0.29]	<i>t</i> = 1.29 {0.312}	0.078 (0.061)	0.066 (0.061)	[−0.09, 0.22] {0.34}
Old functions	3.12 [0.97]	2.76 [1.26]	<i>t</i> = 11.23 {0.241}	0.363 (0.296)	0.345 (0.259)	[−0.15, 1.23] {0.11}
Police force (<i>maréchaussée</i>)	4.16 [1.40]	4.07 [1.86]	<i>t</i> = 0.23 {0.849}	0.093 (0.405)	0.155 (0.405)	[−0.50, 1.59] {0.26}
College (<i>collège</i>)	0.92 [0.28]	0.87 [0.30]	<i>t</i> = 0.56 {0.657}	0.048 (0.086)	0.042 (0.091)	[−0.11, 0.25] {0.51}
<i>Panel C. Economic potential</i>						
Wheat suitability (GAEZ)	6.23 [1.08]	6.35 [1.07]	<i>t</i> = −0.73 {0.477}	−0.124 (0.170)	−0.077 (0.182)	[−0.41, 0.61] {0.69}
Caloric yield	1.89 [0.36]	1.89 [0.38]	<i>t</i> = −0.17 {0.865}	−0.004 (0.025)	−0.005 (0.026)	[−0.21, 0.04] {0.43}
Salt office (<i>bureau des traites</i>)	0.40 [0.50]	0.34 [0.44]	<i>t</i> = 0.71 {0.528}	0.055 (0.078)	0.032 (0.081)	[−0.13, 0.19] {0.58}
Salt granary (<i>grenier à sel</i>)	0.20 [0.41]	0.21 [0.38]	<i>t</i> = −0.23 {1.00}	−0.015 (0.066)	−0.028 (0.068)	[−0.17, 0.11] {0.89}
Height of conscripts	170.65 [2.85]	169.61 [2.63]	<i>t</i> = 1.16 {0.261}	0.843 (0.698)	0.890 (0.701)	[−0.98, 2.22] {0.52}

Notes: This table reports baseline characteristics for our augmented sample of 25 departments measured in capital cities (column 1) compared to other candidate cities (column 2). Standard deviations are reported in brackets. Column 3 reports the *t*-statistic from a paired *t*-test of equality of means between capitals and other candidates, with $N - 1$ degrees of freedom, where $N = 25$ is the number of departments. In curly brackets, we report the randomization inference (RI) *p*-value based on this *t*-stat obtained after 10,000 permutations of the treatment. Column 4 reports the average difference between capitals and other candidates. Column 5 reports the coefficient from regressions of each variable on capital status, where each regression includes department fixed effects (equation (1)). Heteroskedasticity-robust standard errors are reported in parentheses. The *F*-stat from the test of joint significance of all coefficients in column 5 is $F = 1.37$ ($p = 0.225$). Column 6 reports 95 percent permutation-based randomization confidence intervals and (in curly brackets) the randomization inference *p*-value testing the significance of the studentized treatment effect reported in column 5, obtained after 50,000 permutations of the treatment. Across all columns, the sample includes 69 cities in 25 departments.

city population in 1793 and 1800 and population growth between 1793 and 1800; administrative functions under the Ancien Régime (dummy variables indicating the presence of a bishopric, bailiwick, tax center, and administrative center, the sum of these functions, as well as the size of the police force and the presence of a secondary school in 1789); and pre-reform measures of economic potential (including GAEZ wheat suitability from Fischer and Nachtergaele 2021 and the potential caloric yield from Galor and Özak 2016), the presence of salt taxation offices (Touzery 2024), and the height of adult conscripts born after 1700 (Komlos 2006).

Columns 1 and 2 of Tables 2 and 3 report averages of these variables across the treatment group (capitals) and the control group (other candidates). Column 3 reports the t -statistic from a paired t -test with $N - 1$ degrees of freedom, where N is equal to the number of departments in the sample (i.e., either 17 or 25). In column 4, we report the average of the within-department treatment-control differences, alongside the standard error of this difference, which is calculated using the variance of the treatment-control difference across departments. Column 5 reports the estimate of α_1 from equation (1), and column 6 reports the randomization inference p -value and the 95 percent randomization-based confidence interval around this estimate, following DiCiccio and Romano (2017) and Young (2024).

Overall, capital cities appear no different from other candidate cities before the reform. Only 1 out of 20 coefficients is significant (at the 5 percent level). Indeed, capital cities are slightly more likely to have a tax center but less likely to have a bishopric or a bailiwick, and they also have lower wheat suitability. Table 7 reports additional balance checks pertaining to the periphery of capitals; columns 1 and 3 report estimates of β_1 from equation (2), and columns 2 and 4 reports randomization inference p -values and randomization confidence intervals in the baseline and the augmented samples, respectively.³⁴ Again, we find no evidence that final capitals differ from other candidates in any substantial way.

Nonetheless, Figure 1 suggests that the final capitals are often more centrally located within departments; on average, capitals are 16 (13) kilometers closer to their department centroid than other candidates in the baseline (augmented) sample. Table 6, where we regress capital status on centrality alongside other predetermined characteristics, shows this formally. We find that capitals are located 45–49 percent closer to the new department centroid, and this difference is statistically significant (column 1). Importantly, we show in Section IVF that this does not affect our results, and this is for two main reasons. First, centrality is defined relative to borders that, for the artificial departments included in our sample, were themselves arbitrary. Since the departments we study were not built around one specific city, centrality inside the new departments would not affect economic development in the absence of the reform.³⁵ Second, the centrality criterion may have played a role because the revolutionaries pursued an ideal of equity; even the most isolated citizens had

³⁴ We do not include pre-reform administrative functions in those checks, as a negligible number of municipalities outside candidate cities have any administrative function. There are no significant differences between the two groups of cities when we look at these functions.

³⁵ Arnould (1992, p. 402) provides relevant background on the arbitrariness of department boundaries: “The members of the Constituent Assembly (. . .) established no link between the existing urban network and the location of future capitals. This indifference betrayed a hidden animosity towards cities (. . .). In general, the settlement of capitals was conceived as a task to be undertaken after constituencies were drawn (. . .). For two centuries, the same has been said about the demarcation of departments: its arbitrary and artificial character.”

to be able to reach the capital in a reasonable amount of time (de Condorcet 1788). However, this equity concern was a priori unrelated to efficiency considerations, such as maximizing the economic output of the capital, and thus unlikely to directly affect our outcomes. In fact, only 11 of the 25 capitals were the city with the highest urban potential, a more natural measure for the economic potential of the city.

Consistent with this, in Tables 2 and 3, we find no correlation between capital status and pre-reform urban potential or market access. Furthermore, the correlation between proximity to the department centroid and pre-reform market access is, as expected, negligible; this correlation is 0.09 when measured in levels and -0.01 when measured in logs. Nonetheless, we show that our main results are robust to controlling for both proximity to the centroid and pre-reform market access and urban potential. Including or excluding these controls leaves our results unchanged.

IV. The Early Buildup of Coercive Capacity

A. Overview

State-building efforts such as the 1790 reform can trigger long-term benefits for the chosen capitals, but these gains can take time to materialize. Figure 2 illustrates the dynamic effects of this reform. In this figure, we plot the regression coefficients from a modified version of equation (1) estimated on a panel dataset at the municipality-year level including all the candidate cities in our baseline sample. Online Appendix Figure A.2 reports the corresponding figures for the augmented sample. As we show across these different panels, the state-building process begins with investments in taxation and law enforcement (panels A and B), which translate in the medium run into higher public goods provision (panel C). In the long run, this permeates to the private sector (panels D and E) and population (panel F). In this section, we estimate the short- and medium-term impacts of the reform, focusing on measures of taxation and law enforcement (Section IVB) as well as the provision of public goods, such as schools and hospitals (Section IVC). We also show that economic benefits were very limited in the short run (Section IVD).

B. Investments in Taxation and Law Enforcement

Tables 4 and 5, panel A examine the short- and medium-run effects of the reform on various measures of investments in taxation and law enforcement, which we refer to as coercive capacity in the model. Tables 8 and 9 extend this analysis to the periphery of capitals and other candidate cities.

Taxation.—The first outcome we consider is the timing of the establishment of municipal cadastres. Cadastres were rolled out across France between 1807 and 1850 to facilitate and harmonize land taxation (see Section ID). The first row of Tables 4 and 5 shows that capitals were approximately 38 percentage points more likely to have a cadastre by the end of the Napoleonic period in 1815. Figure 2, panel A shows this dynamic evolution of the cadastre in capital cities, relative to other candidates. By 1830, capitals remain substantially more likely to have a cadastre, but this differential effect vanishes by the middle of the nineteenth century, at which

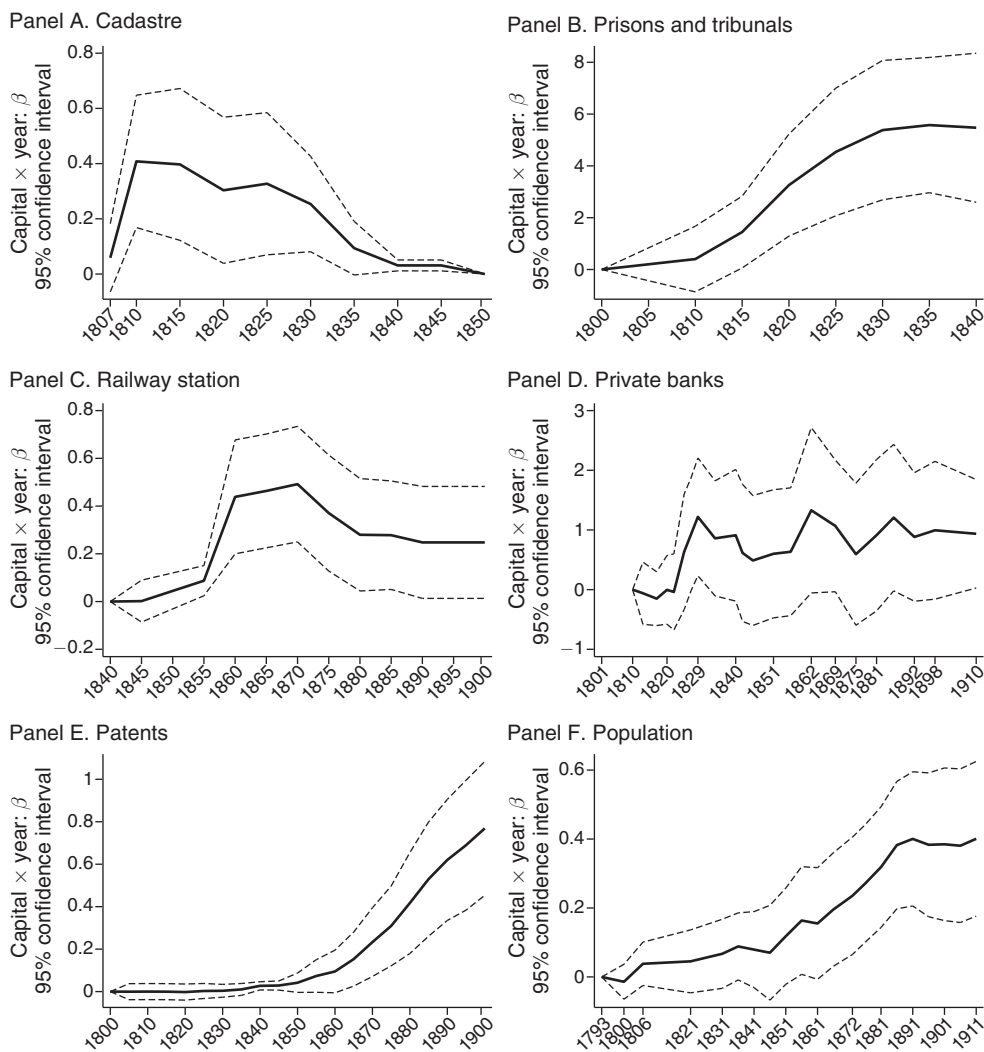


FIGURE 2. DYNAMIC EFFECTS OF CAPITAL STATUS

Notes: This figure shows the dynamic effects of the administrative reform over the nineteenth century. We plot the regression coefficients from a modified version of equation (1) estimated on a panel dataset at the municipality-year level, including municipality and year fixed effects. The sample includes the 50 candidate cities in our baseline sample. The dependent variable is in panel A, a dummy variable equal to 1 if a municipality has established a cadastre by year t ; in panel B, the cumulative number of construction and renovation projects of prisons and tribunals by year t ; in panel C, a dummy variable equal to 1 if the municipality has a railway station by year t ; in panel D, the number of banks operating in the municipality in year t (we drop the year 1864 due to data quality issues); in panel E, the number of patents registered by residents of the municipality in year t per 100 inhabitants in 1793; in panel F, log population measured in year t . The dashed lines indicate 95 percent confidence intervals, with standard errors two-way clustered by municipality and by department-year.

point nearly all municipalities have been covered by the cadastral effort. These effects extend to towns and villages in the periphery of capitals; nearby municipalities are also more likely to have their cadastre established by 1815 (Table 8 and Table 9, columns 1–3).

TABLE 4—EFFECTS OF CAPITAL STATUS: BASELINE SAMPLE

Timing	Outcome	Capitals [SD] (1)	Candidates [SD] (2)	<i>t</i> -test {RI <i>p</i> -value} (3)	Difference (SE) (4)	OLS (SE) (5)	Rand. inference [95% CI] and { <i>p</i> -value} (6)
<i>Panel A. Taxation and enforcement</i>							
Short	Cadastre by 1815	0.65 [0.49]	0.27 [0.39]	<i>t</i> = 2.61 {0.027}	0.375 (0.144)	0.385 (0.140)	[0.12, 0.67] {0.00}
	Business tax collected, 1839	3.39 [4.78]	1.24 [1.11]	<i>t</i> = 1.89 {0.057}	2.152 (1.136)	2.582 (1.299)	[−0.59, 5.10] {0.18}
	Conscripts 1802–1815	5.12 [0.51]	4.97 [0.56]	<i>t</i> = 1.40 {0.183}	0.153 (0.110)	0.176 (0.120)	[−0.04, 0.53] {0.09}
	Police force in 1816	3.18 [0.81]	0.91 [0.51]	<i>t</i> = 7.69 {0.000}	2.262 (0.294)	2.329 (0.315)	[1.84, 2.90] {0.00}
	Tribunal project by 1815	0.71 [0.47]	0.38 [0.45]	<i>t</i> = 2.68 {0.032}	0.324 (0.121)	0.305 (0.118)	[0.10, 0.78] {0.02}
<i>Panel B. Public goods</i>							
Short	Secondary school, 1812	0.94 [0.24]	0.79 [0.37]	<i>t</i> = 1.40 {0.262}	0.147 (0.105)	0.139 (0.119)	[0.00, 0.48] {0.09}
	Hospital project by 1815	0.35 [0.49]	0.30 [0.43]	<i>t</i> = 0.29 {0.863}	0.051 (0.176)	0.022 (0.138)	[−0.23, 0.48] {0.50}
	Secondary school, 1836	0.94 [0.24]	0.78 [0.37]	<i>t</i> = 1.65 {0.180}	0.162 (0.098)	0.158 (0.108)	[0.00, 0.52] {0.06}
	Hospital project by 1840	0.65 [0.49]	0.39 [0.44]	<i>t</i> = 1.82 {0.107}	0.260 (0.143)	0.201 (0.136)	[0.00, 0.68] {0.06}
Long	Telegraph connection, 1863	0.88 [0.33]	0.04 [0.12]	<i>t</i> = 8.25 {0.000}	0.846 (0.103)	0.841 (0.097)	[0.64, 1.00] {0.00}
	Welfare beneficiaries, 1871	10.77 [4.62]	8.12 [3.98]	<i>t</i> = 2.37 {0.031}	2.649 (1.117)	2.584 (1.140)	[0.25, 6.71] {0.04}
	Train station, 1870	0.82 [0.39]	0.48 [0.47]	<i>t</i> = 3.20 {0.014}	0.346 (0.108)	0.362 (0.102)	[0.17, 0.57] {0.00}

(continued)

While intuitive and consistent with our conceptual framework, this early buildup of the ability to raise taxes via the establishment of cadastres may not have occurred had the government pursued different objectives. Indeed, a government seeking to maximize rents under the constraint of avoiding civil unrest may have sought to build up fiscal capacity further away from the capital (as in Campante and Do 2014). Instead, our results show that the modern French state first projected its fiscal powers in and around the new local capitals, before extending this tax collection capacity to the rest of the territory.³⁶

³⁶ Nonetheless, in addition to facilitating taxation, the cadastre could also have had positive effects via its impact on, e.g., efficient litigation or property rights enforcement. The effects we show below suggest that these potential benefits did not translate into faster growth in the short run. Moreover, online Appendix Table A.4, using data from the Historical Social Conflict Database (Chambru and Maneuvrier-Hervieu 2024), compares the number of riots between 1800 and 1848 in the capital versus other candidates. We find no significant effect of capital status on conflicts, regardless of their nature (fiscal riots, riots against the state, or other types of riots). This suggests that short-term positive effects of the cadastre on efficient litigation were likely limited, as they did not reduce the frequency of conflicts.

TABLE 4—EFFECTS OF CAPITAL STATUS: BASELINE SAMPLE (*continued*)

Timing	Outcome	Capitals [SD] (1)	Candidates [SD] (2)	<i>t</i> -test {RI <i>p</i> -value} (3)	Difference (SE) (4)	OLS (SE) (5)	Rand. inference [95% CI] and { <i>p</i> -value} (6)
<i>Panel C. Economic effects</i>							
Short	Industrial establishments, 1839	0.22 [0.28]	0.24 [0.58]	$t = -0.12$ {0.979}	-0.014 (0.115)	-0.004 (0.179)	[-0.42, 0.26] {0.74}
	Output per worker, 1839	4.77 [3.63]	5.62 [4.50]	$t = -0.13$ {0.900}	-0.249 (1.599)	-0.284 (2.164)	[-3.40, 3.12] {0.88}
	log population, 1846	9.17 [0.41]	8.99 [0.48]	$t = 1.46$ {0.161}	0.179 (0.123)	0.195 (0.126)	[-0.05, 0.58] {0.09}
	Private banks, 1851	2.76 [1.44]	2.67 [1.40]	$t = 0.19$ {0.861}	0.093 (0.490)	0.258 (0.534)	[-0.47, 1.83] {0.26}
	Patents registered by 1850	0.14 [0.11]	0.11 [0.13]	$t = 0.90$ {0.374}	0.030 (0.033)	0.038 (0.033)	[-0.02, 0.11] {0.18}
Long	Private banks, 1910	3.71 [1.36]	2.81 [1.34]	$t = 1.93$ {0.076}	0.895 (0.464)	0.945 (0.491)	[0.16, 1.94] {0.02}
	Patents registered by 1900	1.29 [0.57]	0.58 [0.48]	$t = 5.05$ {0.000}	0.715 (0.142)	0.770 (0.139)	[0.51, 1.06] {0.00}
	log population, 1886	9.59 [0.43]	9.12 [0.51]	$t = 3.54$ {0.002}	0.462 (0.130)	0.466 (0.136)	[0.18, 0.95] {0.01}
	log pop., no civil servants, 1886	9.39 [0.59]	9.05 [0.53]	$t = 2.09$ {0.051}	0.341 (0.163)	0.302 (0.185)	[-0.05, 0.82] {0.08}
	log population, 1999	10.11 [0.55]	9.45 [0.71]	$t = 3.91$ {0.002}	0.663 (0.169)	0.636 (0.184)	[0.22, 1.02] {0.00}
	Private employees (p.c.), 2015	0.62 [0.15]	0.42 [0.09]	$t = 4.87$ {0.000}	0.195 (0.040)	0.197 (0.037)	[0.12, 0.28] {0.00}

Notes: This table reports endline outcomes for our baseline sample of 17 departments measured in capital cities compared to other candidate cities. Standard deviations are reported in brackets. Column 3 reports the *t*-statistic from a paired *t*-test of equality of means between capitals and other candidates, with $N - 1$ degrees of freedom, where $N = 17$ is the number of departments. In curly brackets, we report the randomization inference (RI) *p*-value based on this *t*-stat obtained after 10,000 permutations of the treatment. Column 4 reports the average difference between capitals and other candidates. Column 5 reports the coefficient from regressions of each variable on capital status, where each regression includes department fixed effects (equation (1)). Heteroskedasticity-robust standard errors are reported in parentheses. Column 6 reports 95 percent permutation-based randomization confidence intervals and (in curly brackets) the randomization inference *p*-value testing the significance of the studentized treatment effect reported in column 5, obtained after 50,000 permutations of the treatment. Across all columns, the sample includes 50 cities across 17 departments.

Consistent with the interpretation of the cadastre as a fiscal instrument, we examine the effects of the reform on the amount of business tax (*patentes*) collected in the country's first industrial census, conducted between 1839 and 1847. Our main outcome of interest is the tax collected per ten inhabitants at baseline (in 1793). The state collects more taxes in capitals than in other candidates; this difference is significant in Table 4 but falls short of statistical significance in Table 5.

Law Enforcement.—We also examine the short-term effects of the reform on police presence and on constructions and renovations of tribunals. By the end of the Napoleonic period, more police forces are stationed in capitals, and more tribunals are built inside these cities, relative to other candidates. This occurs even though

TABLE 5—EFFECTS OF CAPITAL STATUS: AUGMENTED SAMPLE

Timing	Outcome	Capitals [SD] (1)	Candidates [SD] (2)	<i>t</i> -test {RI <i>p</i> -value} (3)	Difference (SE) (4)	OLS (SE) (5)	Rand. inference [95% CI] and { <i>p</i> -value} (6)
<i>Panel A. Taxation and enforcement</i>							
Short	Cadastral by 1815	0.63 [0.49]	0.24 [0.38]	<i>t</i> = 3.33 {0.005}	0.377 (0.111)	0.384 (0.112)	[0.15, 0.60] {0.00}
	Business tax collected, 1839	2.83 [4.12]	2.10 [2.81]	<i>t</i> = 0.69 {0.512}	0.737 (1.068)	0.907 (1.265)	[−1.78, 2.94] {0.76}
	Conscripts 1802–1815	5.19 [0.52]	5.02 [0.50]	<i>t</i> = 1.74 {0.096}	0.167 (0.096)	0.181 (0.104)	[−0.01, 0.48] {0.06}
	Police force in 1816	3.16 [1.03]	0.91 [0.49]	<i>t</i> = 8.06 {0.000}	2.252 (0.279)	2.246 (0.315)	[1.74, 2.81] {0.00}
	Tribunal project by 1815	0.68 [0.48]	0.31 [0.43]	<i>t</i> = 3.08 {0.007}	0.367 (0.119)	0.344 (0.113)	[0.15, 0.72] {0.00}
<i>Panel B. Public goods</i>							
Short	Secondary school, 1812	0.92 [0.28]	0.75 [0.39]	<i>t</i> = 1.84 {0.074}	0.173 (0.094)	0.154 (0.105)	[0.00, 0.44] {0.05}
	Hospital project by 1815	0.36 [0.49]	0.28 [0.43]	<i>t</i> = 0.54 {0.667}	0.075 (0.138)	0.061 (0.114)	[−0.14, 0.43] {0.39}
	Secondary school, 1836	0.92 [0.28]	0.83 [0.33]	<i>t</i> = 1.09 {0.376}	0.090 (0.083)	0.081 (0.099)	[−0.09, 0.43] {0.19}
	Hospital project by 1840	0.68 [0.48]	0.38 [0.46]	<i>t</i> = 2.36 {0.030}	0.297 (0.126)	0.254 (0.118)	[0.06, 0.68] {0.02}
Long	Telegraph connection, 1863	0.88 [0.33]	0.04 [0.12]	<i>t</i> = 9.81 {0.000}	0.842 (0.086)	0.820 (0.093)	[0.64, 0.96] {0.00}
	Welfare beneficiaries, 1871	11.65 [6.26]	8.04 [4.36]	<i>t</i> = 2.94 {0.005}	3.608 (1.228)	3.564 (1.167)	[1.30, 7.55] {0.01}
	Train station, 1870	0.72 [0.46]	0.42 [0.46]	<i>t</i> = 2.88 {0.012}	0.295 (0.102)	0.292 (0.107)	[0.08, 0.46] {0.01}

(continued)

there was no specific law regarding the location of judicial courts.³⁷ This gap in enforcement capacity continues to grow until 1840; Figure 2, panel B, where we jointly look at tribunals and prisons, shows the time path of this divergence until 1840. These effects illustrate the importance given to law enforcement institutions in the early buildup of state capacity after the Revolution.

Conscription.—Extractive capacity implies the ability to collect not only taxes but also physical resources from citizens, in particular during conflicts. Throughout the Napoleonic wars, the recruitment of conscripts was placed under the direct responsibility of the department prefects. The conscription system was introduced in 1798

³⁷For instance, the *cours d'assises* (criminal courts involving a jury trial) were often built in the department capital, but this was neither mandated by law nor systematically the case. In 13 departments, the court of appeal was located outside the capital. Commercial courts were more numerous (numbering more than 200) and usually located in industrial and commercial cities (Joanne 1864, pp. CXLIX–CLII).

TABLE 5—EFFECTS OF CAPITAL STATUS: AUGMENTED SAMPLE (*continued*)

Timing	Outcome	Capitals [SD] (1)	Candidates [SD] (2)	<i>t</i> -test {RI <i>p</i> -value} (3)	Difference (SE) (4)	OLS (SE) (5)	Rand. inference [95% CI] and { <i>p</i> -value} (6)
<i>Panel C. Economic effects</i>							
Short	Industrial establishments, 1839	0.19 [0.24]	0.28 [0.53]	$t = -0.98$ {0.362}	-0.095 (0.096)	-0.094 (0.138)	[-0.41, 0.12] {0.36}
	Output per worker, 1839	5.20 [4.08]	5.42 [4.68]	$t = 0.19$ {0.856}	0.313 (1.395)	0.276 (1.755)	[-2.50, 3.38] {0.93}
	log population, 1846	9.18 [0.42]	9.01 [0.42]	$t = 1.68$ {0.109}	0.176 (0.105)	0.183 (0.112)	[-0.02, 0.49] {0.07}
	Private banks, 1851	2.52 [1.42]	2.43 [1.76]	$t = 0.19$ {0.867}	0.090 (0.474)	0.088 (0.559)	[-0.58, 1.02] {0.57}
	Patents registered by 1850	0.15 [0.12]	0.11 [0.12]	$t = 1.58$ {0.123}	0.039 (0.024)	0.046 (0.025)	[-0.01, 0.09] {0.10}
Long	Private banks, 1910	4.00 [1.96]	2.94 [1.29]	$t = 2.36$ {0.028}	1.062 (0.450)	0.990 (0.483)	[0.20, 1.99] {0.02}
	Patents registered by 1900	1.29 [0.61]	0.65 [0.68]	$t = 5.14$ {0.000}	0.635 (0.124)	0.659 (0.153)	[0.30, 0.91] {0.00}
	log population, 1886	9.58 [0.43]	9.18 [0.44]	$t = 3.68$ {0.001}	0.399 (0.108)	0.402 (0.120)	[0.17, 0.80] {0.00}
	log pop., no civil servants, 1886	9.38 [0.57]	9.09 [0.47]	$t = 2.18$ {0.039}	0.287 (0.132)	0.256 (0.154)	[-0.02, 0.66] {0.06}
	log population, 1999	10.25 [0.59]	9.59 [0.69]	$t = 4.81$ {0.000}	0.676 (0.138)	0.668 (0.156)	[0.34, 0.98] {0.00}
	Private employees (p.c.), 2015	0.62 [0.14]	0.44 [0.10]	$t = 5.02$ {0.000}	0.173 (0.034)	0.174 (0.033)	[0.11, 0.27] {0.00}

Notes: This table reports endline outcomes for our augmented sample of 25 departments measured in capital cities compared to other candidate cities. Standard deviations are reported in brackets. Column 3 reports the *t*-statistic from a paired *t*-test of equality of means between capitals and other candidates, with $N - 1$ degrees of freedom, where $N = 25$ is the number of departments. In curly brackets, we report the randomization inference (RI) *p*-value based on this *t*-stat obtained after 10,000 permutations of the treatment. Column 4 reports the average difference between capitals and other candidates. Column 5 reports the coefficient from regressions of each variable on capital status, where each regression includes department fixed effects (equation (1)). Heteroskedasticity-robust standard errors are reported in parentheses. Column 6 reports 95 percent permutation-based randomization confidence intervals and (in curly brackets) the randomization inference *p*-value testing the significance of the studentized treatment effect reported in column 5, obtained after 50,000 permutations. Across all columns, the sample includes 69 cities across 25 departments.

and maintained until the end of the Empire in 1815.³⁸ Tables 4 and 5, panel A show that this system led to higher conscription for capitals. The dependent variable is the log number of conscripts in each city. Overall, capitals enroll approximately 15 percent more men than other candidate cities in the baseline sample and 17 percent in the augmented sample; this effect is significant at 10 percent in the latter (Table 5) but falls slightly short of statistical significance in the baseline sample (Table 4).

³⁸The draft was viewed with fear in rural communities. Forrest (2012, p. 50) points out that the “achievement of conscription targets would be one of the principal gauges of loyalty and efficiency in the Consulate and Empire, a yardstick by which mayors, sub-prefects and prefects would be judged by the Napoleonic state.”

TABLE 6—THE ROLE OF CENTRALITY

	Capital	Cadastre by 1815		Train station 1870		Patents 1900	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A. Baseline sample</i>							
Capital		0.3688 (0.1449)	0.4977 (0.2302)	0.3799 (0.1334)	0.4773 (0.2299)	0.6158 (0.1475)	0.7509 (0.2034)
Centrality	0.4930 (0.0801)		−0.1056 (0.1436)		−0.0799 (0.1303)		−0.1109 (0.1111)
Cities	50	49	49	50	50	50	50
Dep. var. mean	0.340	0.388	0.388	0.580	0.580	0.788	0.788
R^2	0.753	0.693	0.700	0.810	0.814	0.878	0.882
<i>Panel B. Augmented sample</i>							
Capital		0.3392 (0.1436)	0.4223 (0.1729)	0.2973 (0.1124)	0.3021 (0.1587)	0.7792 (0.1725)	0.5882 (0.2334)
Centrality	0.4485 (0.0779)		−0.0940 (0.1155)		−0.0054 (0.0938)		0.2200 (0.2165)
Cities	69	66	66	69	69	69	69
Dep. var. mean	0.362	0.394	0.394	0.551	0.551	0.915	0.915
R^2	0.542	0.640	0.648	0.783	0.783	0.750	0.763
		Banks 1898		Population 1886		Population 1999	
		(8)	(9)	(10)	(11)	(12)	(13)
<i>Panel A. Baseline sample (continued)</i>							
Capital		1.6193 (0.4941)	2.2161 (0.9152)	0.4253 (0.1034)	0.3288 (0.1366)	0.6455 (0.1360)	0.7954 (0.2397)
Centrality			−0.4898 (0.5133)		0.0792 (0.0926)		−0.1230 (0.1331)
Cities		50	50	50	50	50	50
Dep. var. mean		3.620	3.620	9.193	9.193	9.712	9.712
R^2		0.878	0.884	0.957	0.959	0.918	0.921
<i>Panel B. Augmented sample (continued)</i>							
Capital		1.3850 (0.5014)	1.2462 (0.6898)	0.4088 (0.0702)	0.3652 (0.0880)	0.7383 (0.1627)	0.8467 (0.2327)
Centrality			0.1598 (0.3957)		0.0502 (0.0851)		−0.1050 (0.1471)
Cities		69	69	69	69	68	68
Dep. var. mean		3.739	3.739	9.258	9.258	9.831	9.831
R^2		0.822	0.823	0.935	0.936	0.850	0.853

Notes: Column 1 reports estimates from a regression of capital status on centrality (defined as minus log distance to the department centroid). Columns 2 through 13 report estimates from equation (1). All regressions control for department fixed effects, land area and land area squared, minimal and maximal altitude, latitude and longitude, log population in 1793 and 1800, four dummy variables indicating the presence of bishoprics, bailiwicks, tax centers, and *subdélégations*, pre-reform market access, and wheat suitability. In odd-numbered columns, we additionally control for centrality. Robust standard errors in parentheses.

C. Short- and Medium-Run Effects on Public Goods

By contrast with the results shown so far, Tables 4 and 5 (panel B, short-run section) show positive but weaker effects of state building on other types of public goods in the short run. We look at the presence of a secondary school by 1812 and

TABLE 7—EFFECTS ON THE PERIPHERY OF CAPITALS: BALANCE CHECKS

	Main sample		Augmented sample	
	Difference (1)	RI (2)	Difference (3)	RI (4)
log market access (low θ)	−0.082 (0.063) {0.25}	[−0.24, 0.09] {0.28}	−0.105 (0.057) {0.12}	[−0.22, 0.02] {0.10}
log market access (high θ)	−0.199 (0.200) {0.37}	[−0.67, 0.38] {0.51}	−0.193 (0.184) {0.34}	[−0.62, 0.20] {0.28}
log urban potential	−0.003 (0.006) {0.71}	[−0.02, 0.02] {0.98}	0.000 (0.007) {0.96}	[−0.01, 0.03] {0.42}
log population in 1793	0.065 (0.051) {0.25}	[−0.04, 0.16] {0.18}	0.040 (0.046) {0.41}	[−0.05, 0.12] {0.39}
Population growth, 1793–1800	−0.008 (0.013) {0.55}	[−0.04, 0.02] {0.65}	−0.007 (0.011) {0.55}	[−0.03, 0.02] {0.73}
Wheat suitability (GAEZ)	−0.390 (0.181) {0.06}	[−0.77, 0.12] {0.16}	−0.183 (0.166) {0.31}	[−0.52, 0.23] {0.45}
Caloric yield	0.000 (0.017) {0.99}	[−0.03, 0.02] {0.63}	−0.017 (0.019) {0.42}	[−0.04, 0.03] {0.77}
Old functions	−0.003 (0.003) {0.40}	[−0.01, 0.00] {0.21}	0.001 (0.003) {0.80}	[−0.01, 0.01] {0.65}
Distance to bishopric	0.162 (0.130) {0.26}	[−0.22, 0.48] {0.47}	0.178 (0.127) {0.20}	[−0.19, 0.46] {0.45}
Distance to bailiwick	0.072 (0.079) {0.40}	[−0.12, 0.29] {0.44}	−0.017 (0.084) {0.85}	[−0.19, 0.17] {0.92}
Distance to tax center	−0.125 (0.077) {0.14}	[−0.33, 0.00] {0.06}	−0.099 (0.068) {0.17}	[−0.28, 0.02] {0.10}
Distance to admin. center	0.023 (0.045) {0.62}	[−0.09, 0.14] {0.54}	−0.005 (0.039) {0.91}	[−0.07, 0.10] {0.70}

Notes: This table compares baseline outcomes between municipalities located in the periphery of capitals and municipalities located in the periphery of other candidate cities. The sample is composed of all the municipalities within a 20 km radius of candidate cities, excluding the candidates themselves. This includes 3,214 municipalities in the baseline sample and 4,539 municipalities in the augmented sample. Columns 1 and 3 report estimates from equation (2). Standard errors clustered by department are reported in parentheses, and wild-bootstrap p -values are reported in curly brackets. Geographic controls (area and area squared, altitude, latitude and longitude) are included throughout. Columns 2 and 4 report 95 percent permutation-based randomization confidence intervals and (in curly brackets) the randomization inference p -value testing the significance of the studentized treatment effect reported in columns 1 and 3, obtained after 50,000 permutations of the treatment.

an indicator for constructions and renovations of hospitals having taken place by 1815. The effects on hospitals and schools often fall short of statistical significance and are small in magnitude relative to the corresponding coefficients for tribunals.

Yet the early buildup of fiscal capacity may have eventually allowed capitals to fund more local public goods in the medium run. Tables 4 and 5, panel B (bottom of short-run section) present the effects of the reform on two later indicators of public

TABLE 8—EFFECTS ON THE PERIPHERY OF CAPITALS: ENDLINE

	Main sample		Augmented sample	
	Difference (1)	RI (2)	Difference (3)	RI (4)
<i>Panel A. Taxation and enforcement</i>				
Cadastre by 1815	0.133 (0.036) {0.00}	[0.06, 0.22] {0.00}	0.170 (0.028) {0.00}	[0.07, 0.23] {0.00}
Business tax collected	0.368 (0.353) {0.41}	[−0.24, 1.23] {0.30}	0.371 (0.298) {0.28}	[−0.30, 0.73] {0.52}
<i>Panel B. Public goods</i>				
Welfare beneficiaries, 1871	0.216 (0.229) {0.40}	[−0.56, 0.44] {0.83}	0.250 (0.291) {0.42}	[−0.54, 0.66] {0.87}
Train station, 1870	0.010 (0.012) {0.47}	[−0.01, 0.05] {0.20}	0.015 (0.010) {0.18}	[−0.01, 0.03] {0.25}
<i>Panel C. Economic effects</i>				
Industrial establishments, 1839	0.008 (0.009) {0.46}	[−0.01, 0.03] {0.32}	0.007 (0.008) {0.35}	[−0.02, 0.02] {0.95}
Output per worker, 1839	−0.260 (1.880) {0.89}	[−3.98, 4.27] {0.92}	−1.771 (1.840) {0.38}	[−5.59, 2.80] {0.60}
log population, 1846	−0.034 (0.015) {0.06}	[−0.08, 0.01] {0.09}	−0.001 (0.022) {0.97}	[−0.06, 0.02] {0.29}
Banks, 1910	−0.018 (0.010) {0.10}	[−0.03, 0.01] {0.17}	−0.007 (0.009) {0.49}	[−0.02, 0.01] {0.49}
Patents registered by 1900	0.032 (0.018) {0.15}	[−0.01, 0.08] {0.12}	0.092 (0.050) {0.06}	[0.01, 0.10] {0.02}
log population, 1886	−0.044 (0.024) {0.10}	[−0.10, −0.01] {0.03}	0.017 (0.041) {0.78}	[−0.08, 0.06] {0.72}
log population, 1999	0.087 (0.088) {0.39}	[−0.12, 0.24] {0.52}	0.254 (0.097) {0.02}	[0.04, 0.38] {0.01}

Notes: This table compares endline outcomes between municipalities located in the periphery of capitals and municipalities located in the periphery of other candidate cities. The sample is composed of all the municipalities within a 20 km radius of candidate cities, excluding the candidates themselves. This includes 3,214 municipalities in the baseline sample and 4,539 municipalities in the augmented sample. Columns 1 and 3 report estimates from equation (2). Standard errors clustered by department are reported in parentheses, and wild-bootstrap p -values are reported in curly brackets. All specifications control for area and area squared, altitude, latitude and longitude), log population in 1793 and 1800, four dummy variables indicating the presence of bishoprics, bailiwicks, tax centers, and *subdélégations*, distance to the department centroid, pre-reform market access, and wheat suitability. Columns 2 and 4 report 95 percent permutation-based randomization confidence intervals and (in curly brackets) the randomization inference p -value testing the significance of the studentized treatment effect reported in columns 1 and 3, obtained after 50,000 permutations of the treatment.

goods provision, still measured before 1850: the availability of a secondary school in 1836 and hospital buildings by 1840. While the effect on secondary schools remains comparable to that of 1812, there is a strong effect on hospitals. Capitals are more

TABLE 9—THE EARLY BUILDUP OF COERCIVE CAPACITY

	Cadastre			Business tax	Conscripts	Police	Prisons	Tribunal
	1815	1830	1850	1839–1847	1802–1815	1816	1815	1815
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Coercive capacity</i>								
Capital in 1800	0.1679	0.0849	0.0030	1.2344	0.1355	2.2829	0.2386	0.4468
	[0.1321]	[0.1078]	[0.0050]	[1.3325]	[0.0682]	[0.2667]	[0.1544]	[0.1320]
	(0.1263)	(0.1079)	(0.0055)	(1.3562)	(0.0581)	(0.3178)	(0.1321)	(0.1048)
	{0.208}	{0.452}	{0.640}	{0.432}	{0.026}	{0.000}	{0.104}	{0.002}
Closest candidate is capital	0.1343	0.1417	−0.0001	0.3919	−0.0268	0.0022	0.0027	−0.0024
	[0.0319]	[0.0460]	[0.0055]	[0.2823]	[0.0394]	[0.0034]	[0.0024]	[0.0019]
	(0.0354)	(0.0614)	(0.0074)	(0.3579)	(0.0452)	(0.0034)	(0.0023)	(0.0020)
	{0.003}	{0.070}	{0.986}	{0.387}	{0.584}	{0.479}	{0.282}	{0.271}
Number of municipalities	3,044	3,044	3,044	3,264	3,221	3,264	3,264	3,264
DV control mean	0.169	0.553	0.992	5.311	1.947	0.002	0.001	0.001
Department FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.146	0.099	0.027	0.044	0.572	0.836	0.436	0.489
	Sec. schools		Hospitals		Welfare	Telegraph	Railway	
	1812	1836	1815	1840	1871	1863	1852	1870
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel B. Public goods provision</i>								
Capital in 1800	0.1728	0.1870	0.0633	0.2752	2.6698	0.8114	0.0770	0.3890
	[0.0880]	[0.0776]	[0.1405]	[0.1394]	[1.7196]	[0.0934]	[0.0780]	[0.1244]
	(0.1081)	(0.1082)	(0.1480)	(0.1284)	(0.9236)	(0.1090)	(0.0586)	(0.1108)
	{0.178}	{0.221}	{0.671}	{0.049}	{0.017}	{0.000}	{0.208}	{0.008}
Closest candidate is capital	−0.0006	−0.0003	−0.0049	−0.0031	0.2247	0.0016	0.0015	0.0109
	[0.0017]	[0.0019]	[0.0024]	[0.0023]	[0.2008]	[0.0009]	[0.0025]	[0.0088]
	(0.0025)	(0.0016)	(0.0026)	(0.0030)	(0.2450)	(0.0009)	(0.0033)	(0.0124)
	{0.833}	{0.850}	{0.041}	{0.341}	{0.402}	{0.076}	{0.707}	{0.444}
Number of municipalities	3,264	3,264	3,264	3,264	3,264	3,264	3,264	3,264
DV control mean	0.002	0.001	0.001	0.002	1.442	0.000	0.002	0.040
Department FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.738	0.810	0.322	0.505	0.161	0.747	0.090	0.182

Notes: This table reports estimates from equation (3). The sample includes all municipalities located within a 20 km radius of candidate cities in 1790 in the main sample. Standard errors clustered by candidate city are reported in brackets. Standard errors clustered by department are reported in parentheses. Wild bootstrap *p*-values are reported in curly brackets. Controls include log population in 1793 and 1800, latitude and longitude, land area and land area squared, minimal and maximal altitude, a dummy for each of the four Ancien Régime administrative functions—*évêchés* (bishoprics), *bailliages* (bailiwicks), *recettes des finances* (tax centers), and *subdélégations*—centrality, pre-reform market access, and wheat suitability in 20 km radius around each municipality. In columns 1–3 of panel A, the date of the first cadastre is unobserved for 220 municipalities (7 percent of the sample).

likely to host a hospital by 1840 (65 percent of capitals have a hospital project as opposed to 39 percent of candidate cities); this effect is significant at the 5 percent level in the augmented sample.

D. No Economic Gains in the Short and Medium Run

The reform did not yield meaningful economic benefits for capitals during the first half century following its adoption, as we show in Tables 4 and 5 (panel C,

short-run section). Using data from the 1839–1847 industrial census, we show that, irrespective of whether we consider the baseline sample of 50 cities or the augmented sample of 69 cities, the number of industrial establishments and output per worker were not significantly different between the capital and other candidate cities. Furthermore, capital cities did not host more banks or register more patents by 1850. Table 8 shows that there are also no short-term effects of the reform on industrial establishments, banks, and patents in the periphery of capitals.

We provide further evidence in online Appendix Tables A.5 and A.6, using additional information from the industrial census. Online Appendix Table A.5 shows that the numbers of (female and male) workers, output per worker, and wages are not significantly different across the two types of cities. As predicted by the model, these similarities in terms of productivity are reflected in price levels. Online Appendix Table A.6 shows that the prices of essential goods (bread, butter, eggs, and wine) collected between 1849 and 1853 are very similar across capitals and other candidate cities. We also compute a standardized price index, calculated as the weighted average of each of the four commodity prices, and a measure of inflation in this price index between 1849–1853 and 1854–1855.³⁹ This measure of inflation is very similar across the two types of cities. Overall, there were no economic gains in the short run. Section V discusses the reform's impacts on economic outcomes over the longer run.

E. The Role of Technological Shocks

The lack of short-run economic impacts of the reform raises questions about the role of technological shocks, which may be complementary with investments in state capacity. Despite the lack of effects on industrial development on average, capitals might have been in a better position to benefit from innovation. To address this question, we explore potential complementarities between technological breakthroughs and capital status at the time of the 1839–1847 industrial census, which roughly coincides with the period when the First Industrial Revolution took hold in France. If time-varying technological shocks were driving our results, we should observe heterogeneous effects during this early phase of industrialization, in which steam power played a major role (see Ridolfi, Salvo, and Weisdorf 2022 for recent evidence).

Thus, we design a simple test looking at interactions between state building and steam technology adoption in online Appendix Table A.7. In our baseline (augmented) sample, 32 percent (36 percent) of cities had at least 1 operational steam engine by 1839, with the number of engines ranging from 1 to 10. We first check that capitals do not adopt steam technology earlier than other candidates (column 1). This validates the approach adopted in the rest of the table: we estimate a modified version of equation (1), interacting capital status with the number of operational steam engines in 1839. We use as dependent variables four proxies

³⁹ Arch. nat., F 20 712 and F 20 713, Tableaux établis par les commissions de statistique cantonales pour la période quinquennale 1849–1853 et pour les années 1854–1855. To construct weights, we follow Allen (2015), who provides tabulations of the components of the consumption basket required to attain 2,100 calories per day in nineteenth-century Great Britain.

for economic development, also used in Tables 4–5: the number of firms, the number of firms per 100 inhabitants and the business tax collected per 10 inhabitants at baseline (in 1793), and output per worker. All specifications deliver a similar insight. Capital status, as in Tables 4–5, does not affect economic output in the medium run, whereas technology adoption positively affects outcomes, with the exception of output per worker. There is no positive interaction between capital status and steam adoption.

This suggests that there were no synergies between state building and local technological shocks at the time of the First Industrial Revolution. While this does not preclude the possibility that such interactions occurred later, the evidence in online Appendix Table A.7 shows that there is no generic relationship between investments in state capacity and major innovations at the core of the period we study. Capitals grow in the long run, as we show in Section V, not because they are better at exploiting technological shocks but because they gradually benefit from the comparative advantage provided by the status of capital.

F. Robustness Checks

Robustness Checks and Alternative Specifications.—The results in Tables 4 and 5 are robust to various robustness checks and alternative specifications. First, we adopt the same design-based perspective as in column 4 of Tables 4 and 5 but consider alternative ways of weighting the data to account for different treatment probabilities across and within departments. Column 1 of online Appendix Tables A.8 and A.9 reports the baseline estimate from Tables 4–5. Column 2 reports a weighted average of the within-department treatment-control differences, using as weights the inverse of the conditional treatment probability inside each department, $(p_{i|j})^{-1}$. Column 3 reports the randomization inference p -value corresponding to this t -statistic. In columns 4 and 5, we consider an alternative weighting scheme using as weights the inverse probability that city i belongs to stratum (department) j , denoted above as p_j . Overall, these approaches yield similar takeaways as our baseline estimation, and they also do not improve efficiency.

Online Appendix Tables A.10 and A.11 report robustness checks on columns 5 and 6 of Tables 4 and 5, respectively. First, we add controls to the baseline equation (1). These controls include geographic controls (land area and land area squared, minimal and maximal altitude, latitude and longitude), demographic controls (log population in 1793 and 1800), four dummy variables indicating the presence of prior administrative functions (bishoprics, bailiwicks, tax centers, and *subdélégations*), log distance to the department centroid, pre-reform market access, and wheat suitability. Column 2 of online Appendix Tables A.10 and A.11 report the corresponding estimates. Column 3 reports the coefficients and standard errors obtained after using the double selection lasso procedure of Belloni, Chernozhukov, and Hansen (2014) to reduce the number of control variables in column 2, imposing that department dummies are always included.

Second, we use standard errors clustered by department instead of heteroskedasticity-robust standard errors in our baseline estimation. In this case, we also report in curly brackets the p -values from a wild bootstrap with 10,000 replications, following Cameron, Gelbach, and Miller (2008), to account

for the small number of departments. Columns 4 and 5 report standard errors clustered by department obtained without and with the set of controls selected by the lasso procedure in column 3, respectively. Finally, column 6 reports 95 percent randomization-based confidence intervals around the estimates in column 5. Overall, these various approaches deliver the same takeaways as the set of baseline results in Tables 4 and 5, with the exception of the result on conscripts and on total business tax collected, though the other results on early investments in tax collection and enforcement (cadastre, police, and tribunals) are very robust.

Table 9 delivers similar takeaways as Tables 4, 5, and 8. We report results from the alternative specification defined in equation (3) estimated in the main sample (online Appendix Table A.14 reports the corresponding results in the augmented sample). The effects on coercive capacity materialize in the short run, unlike those on public goods provision, which unfold in the subsequent decades. Online Appendix Tables A.16 and A.17 report robustness checks on the periphery results in Table 8. In those tables, we use the same estimation strategy as in the baseline equation (2) but consider instead all the municipalities located within $k = 10$ km or $k = 30$ km of candidate cities (instead of $k = 20$ km in Table 8).

Robustness: The Role of Centrality.—As described above, the only dimension that differentiates capitals from other candidate cities is their degree of centrality within the new departments created in 1790. In this section, we show that centrality does not affect our results. This can first be shown directly by including centrality as a control in all our specifications, which we do in online Appendix Tables A.10 and A.11, as described above. Table 6 shows that, not only does centrality not affect the effect of the capital status, but it is also uncorrelated with our main outcomes of interest. This result holds in online Appendix Table A.12, where we consider alternative specifications controlling for either a third-order polynomial in centrality (panels A and B) or a dummy equal to 1 if city i is the most central candidate (panels C and D).

To further explore the role of centrality, we then exploit variation in the extent to which artificial departments were constrained in their design by the presence of contiguous nonartificial departments. Recall that the reform involved drawing both nonartificial departments (typically named after and centered around one major city) and artificial departments.⁴⁰ In the latter, picking a central city as capital would have been more difficult if borders were exogenously imposed, that is, when the department was entirely surrounded by nonartificial departments. Building on this intuition, we construct a measure of border exogeneity for each artificial department, defined as the percentage of the department border that is contiguous with a nonartificial department.⁴¹ We then reproduce our main results in online Appendix Table A.13, adding to our specification an interaction between capital status and this measure of border exogeneity. The coefficients on this interaction

⁴⁰ Online Appendix Table A.3 shows that the capitals are 60 percent closer to the department centroid in nonartificial departments compared to artificial departments.

⁴¹ This measure appears to capture meaningful variation in the quality of the design of each department: the share of borders contiguous to artificial departments is significantly correlated with the distance between the final capital and population centers (online Appendix Figure A.3).

are all nonsignificant and small in magnitude. This evidence suggests that, even in departments that had more flexibility to design their borders so as to make potential capitals more central, this did not translate in economic advantages for these cities. This further supports the notion that differences in centrality are not a concern for our empirical strategy.

V. The Gradual Dividends of State Building

A. Overview

We now turn to our exploration of the effects of the reform over the long run. This section sheds light on the time horizon over which capitals and their periphery reap the economic dividends from state building. We first examine impacts on public goods and economic development (Section VB) before specifically studying population (Sections VC and VD) as well as contemporary outcomes (Section VE).

B. Public Goods and Economic Outcomes in the Long Run

While the buildup of public goods provision was initially slow, Tables 4 and 5 (panel B, long-run section) show that the divergence between capitals and other candidates fully materializes in the second half of the nineteenth century. First, we report effects of the reform on welfare services enumerated in 1871. Capitals have a significantly higher number of welfare beneficiaries scaled by baseline population, using data from Bucquet (1874). Second, we explore the impact of capital status on communication and transportation infrastructure. Capitals are substantially more likely to have a national telegraph connection allowing them direct communication with Paris, and more likely to have a train station by 1870. Approximately half of the candidate cities not chosen as capitals have a train station by this date, and this figure is 35 percentage points higher for capitals. Figure 2, panel C shows dynamic effects on the construction of train stations starting in 1840.

We highlight that some of these effects can be considered a direct by-product of the status of capital. For instance, the telegraph was explicitly set up by an 1852 law to deliver governmental instructions to the whole territory and therefore had to connect department capitals. Other effects on public goods provision are less likely to derive mechanically from administrative presence; investments in the railway network, for example, were mainly shouldered by private investors and aimed to connect major economic hubs as much as administrative centers. Regardless of the origin of these investments, they can be seen as a factor facilitating the growth of capitals since firms and citizens could directly or indirectly benefit from these infrastructure improvements. The telegraph, for example, was key for the early growth of banking during the period we study (Lin et al. 2021).

The buildup of public goods provision eventually delivers substantial benefits for the private sector (Tables 4–5, panel C, long-run section). By 1900, the citizens residing in capitals have registered twice the number of patents registered in other candidate cities. Table 8 shows that this effect extends to municipalities in the near periphery of capitals, relative to municipalities in the vicinity of other candidates. Capitals also outperform the latter cities in terms of financial development, as they

host a larger number of private banks by 1900.⁴² Figure 2, panels D and E display the time paths of effects on private banks and innovation; we look at the number of banks in regular time intervals between 1810 and 1910 and at the number of patents registered by capital residents between 1800 and 1900, respectively. The divergence between capitals and other candidates fully materializes after 1850 along both these dimensions.

C. Population Growth

The gradual acceleration of economic development might eventually translate into urban growth. We thus study the evolution of population size, which—like innovation and financial development—we can track finely over time. We look at log population computed from decennial censuses.

Figure 2, panel F reports dynamic estimates of the demographic impacts of the reform, and Tables 4–5, panel C summarize effects in the long run. The demographic divergence between capitals and other candidates fully materializes from the 1860s onward. The magnitude of this effect becomes substantial by the end of the nineteenth century; on the eve of World War I, capitals are more than 40 percent more populated than their 1790 rivals. Only about a third of this difference is explained by the presence of civil servants and their families, as described in Section VD. As we discuss in Section VE, this divergence continues throughout the twentieth century and eventually spills over to the near periphery of capital cities (Table 8).

Figures 3 and 4 further illustrate the long-term demographic divergence of capitals. In Figure 3, we plot the raw population data across all candidate cities in six of our departments of interest. The figure illustrates that capitals do not outgrow their rivals in the short run but overtake their rivals during the second half of the nineteenth century. Figure 4 depicts the aggregate effect in the left panels, both for the main and augmented samples. We plot log population after residualizing department fixed effects and baseline population (measured in 1793). There is a small level difference that remains constant until approximately 1860, which for the most part can be explained by the relative increase in the number of civil servants and their families residing in capitals, as we show below.

D. Accounting for Civil Servants

How much of the population growth experienced by capitals is driven by civil servants and their families? We collect additional data and present a simple empirical exercise to quantify the share of the population growth that can be attributed to civil servants and their families.

First, we derive information on the total population of civil servants and their families across all departments in our sample from the 1886 population census.⁴³ By this date, the population of capitals is 40 percent to 46 percent higher than the

⁴² By 1910, 8 percent of municipalities in our sample have a local bank branch, with the number of branches varying between 1 and 13. The data on local banking activity span 1801 to 1910, but only two cities in our sample have a bank branch in 1801.

⁴³ Bib. nat. fr., Résultat du dénombrement de 1886. 1ère partie. Paris, Berger-Levrault, pp. 256–316, <https://gallica.bnf.fr/ark:/12148/bc6p06xrtrb>.

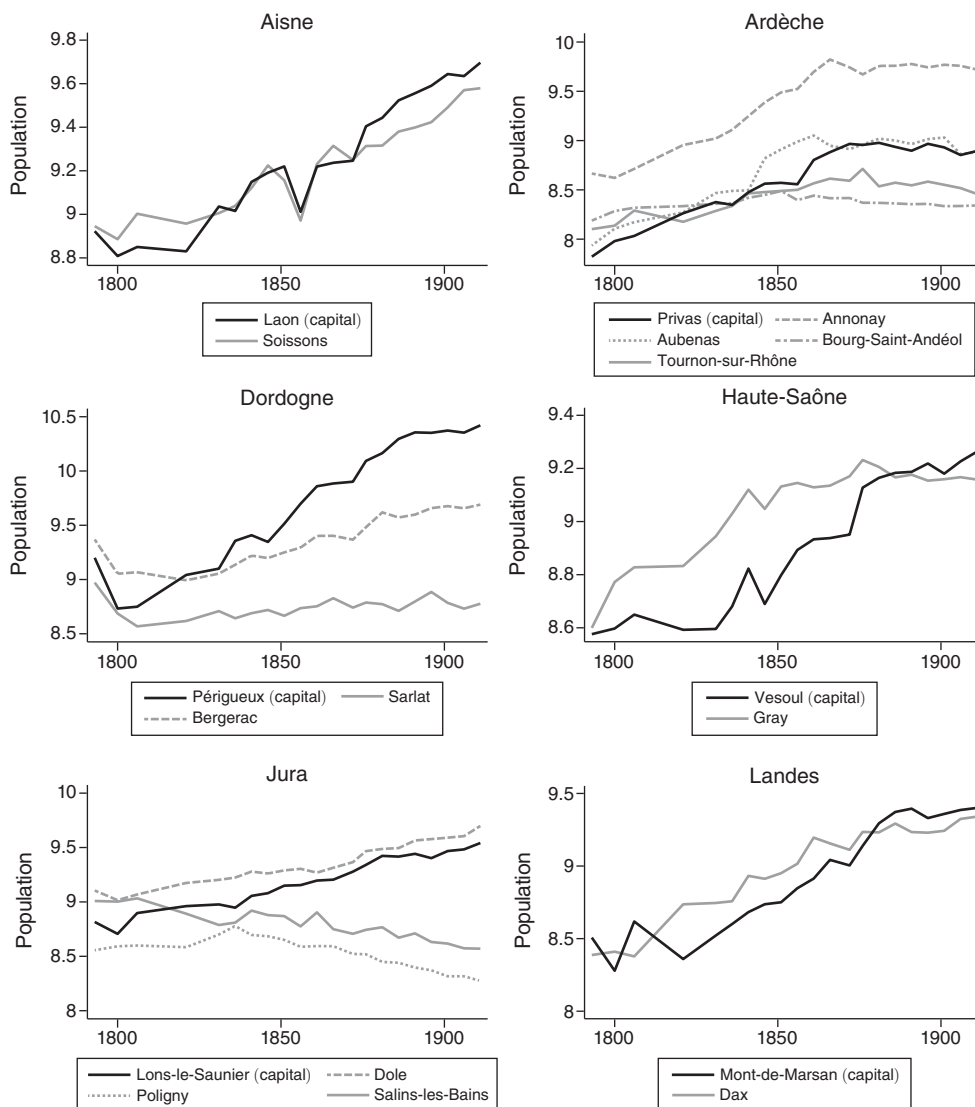
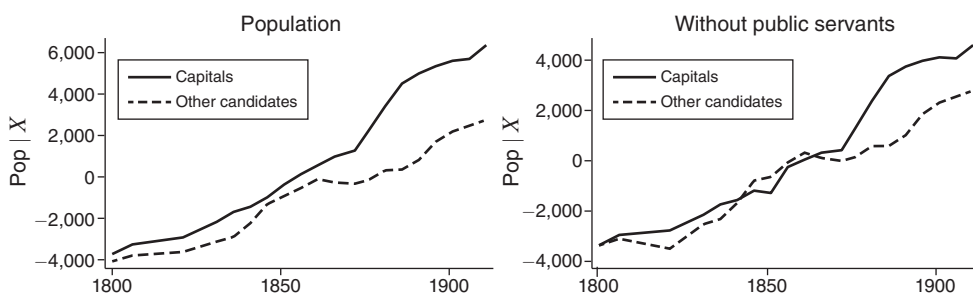


FIGURE 3. DEMOGRAPHIC DIVERGENCE OF CAPITALS: EXAMPLES

Note: In this figure, we plot the raw population (in logs) of capital cities and other candidate cities between 1793 and 1911 in six artificially created departments: Aisne, Ardèche, Dordogne, Haute-Saône, Jura, and Landes.

population of candidate cities across the augmented and baseline samples, respectively. Second, we collect individual data on local civil servants from three departments (Cantal, Jura, and Tarn) to estimate the fraction of all civil servants who resided in the capital, candidate cities, and other municipalities in 1886. Across these departments, the average fraction of a department's civil servants located in the capital is 19 percent. The corresponding fraction for all other candidate cities is 10 percent. Civil servants located in the capital tend to be employed in the local administration (*administration préfectorale*), secondary education, judicial functions, and civil engineering. Civil servants residing outside the capital tend to work in primary instruction, security forces, and postal offices.

Panel A. Main sample



Panel B. Augmented sample

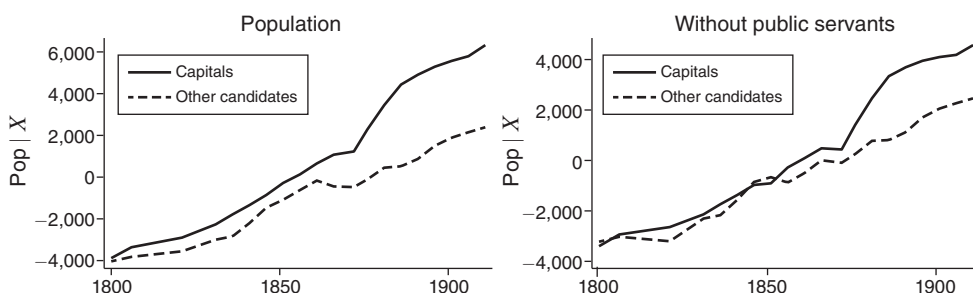


FIGURE 4. DEMOGRAPHIC DIVERGENCE OF CAPITALS

Notes: This figure displays the evolution of population for capitals and other candidate cities between 1793 and 1911. Panel A looks at the baseline sample (50 cities), and panel B looks at the augmented sample (69 cities). We residualize department fixed effects and 1793 population. In the left-hand-side panels, we use total population, while on the right-hand side, we use population net of the estimated number of public servants and their families (see Section IVD).

We then subtract from the population of each city the estimated number of civil servants with their families. For the capital, this estimate corresponds to 19 percent of the total number of civil servants in the department, using the figure obtained above. For other candidate cities, this estimate is equal to 10 percent of the same total, divided by the number of candidate cities.⁴⁴ We allocate equally all remaining civil servants across all other municipalities. Finally, we estimate again equations (1) and (3), using as our dependent variable the estimated (log) city population excluding civil servants and their families.

After implementing this correction, the effect of capital status on 1886 population remains large and statistically significant. In Tables 4 and 5, panel C, the effect size falls to approximately 27–35 percent after removing civil servants and their families. In column 6 of Table 10, the point estimate is also two-thirds of the size of the estimate obtained in column 5, where we look at 1886 population including civil servants. Overall, this exercise implies that civil servants and their families contributed about a third to the higher population growth experienced by local administrative centers by the end of the nineteenth century.

We extend this method in Figure 4, where we plot the evolution of population net of the estimated population of civil servants and their families. To do this, we

⁴⁴This denominator is capped at four, the largest number of candidate cities we observe in this exercise (in Jura).

TABLE 10—DYNAMIC EFFECTS ON ECONOMIC DEVELOPMENT

	Establishments	Output/worker	Private banks			Patents		
	1839–1847	1839–1847	1851	1869	1910	1850	1870	1914
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Effects on private sector activity</i>								
Capital in 1800	−0.0783 [0.1324] (0.1009) {0.574}	0.9923 [2.0111] (1.5201) {0.496}	0.6216 [0.4207] (0.5808) {0.354}	1.1407 [0.4682] (0.3744) {0.017}	1.0246 [0.4137] (0.4681) {0.059}	0.0304 [0.0398] (0.0291) {0.341}	0.1990 [0.0933] (0.0676) {0.022}	0.7218 [0.1718] (0.1378) {0.001}
Closest candidate is capital	0.0085 [0.0082] (0.0088) {0.400}	−0.3504 [1.2353] (1.6253) {0.835}	−0.0000 [0.0060] (0.0068) {0.999}	0.0042 [0.0083] (0.0090) {0.654}	−0.0063 [0.0104] (0.0140) {0.765}	0.0085 [0.0040] (0.0055) {0.111}	0.0175 [0.0084] (0.0116) {0.189}	0.0331 [0.0130] (0.0175) {0.116}
Number of municipalities	3,264	310	3,264	3,264	3,264	3,264	3,264	3,264
DV control mean	0.020	6.038	0.002	0.014	0.026	0.009	0.032	0.060
Department FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.078	0.293	0.718	0.664	0.745	0.019	0.042	0.112
log population in:								
	1800	1821	1846	1866	1886	1886	1911	1999
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel B. Effects on population</i>								
Capital in 1800	−0.0105 [0.0445] (0.0366) {0.766}	0.0683 [0.0427] (0.0463) {0.163}	0.1188 [0.0684] (0.0608) {0.078}	0.2736 [0.0848] (0.0691) {0.002}	0.4386 [0.0941] (0.0844) {0.000}	0.2922 [0.1114] (0.1129) {0.017}	0.4630 [0.1121] (0.1019) {0.002}	0.7199 [0.1505] (0.1474) {0.001}
Closest candidate is capital	−0.0102 [0.0107] (0.0128) {0.439}	−0.0091 [0.0070] (0.0087) {0.355}	−0.0325 [0.0121] (0.0151) {0.070}	−0.0376 [0.0127] (0.0163) {0.053}	−0.0423 [0.0179] (0.0234) {0.114}	−0.0497 [0.0192] (0.0253) {0.085}	−0.0473 [0.0285] (0.0390) {0.288}	0.0866 [0.0674] (0.0864) {0.384}
Number of municipalities	3,264	3,254	3,261	3,261	3,264	3,260	3,264	3,076
DV control mean	6.075	6.179	6.290	6.235	6.148	6.089	5.982	5.785
Department FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.945	0.963	0.937	0.923	0.892	0.877	0.843	0.703

Notes: This table reports estimates from equation (3). The sample includes all municipalities located within a 20 km radius of candidate cities in 1790 in the main sample. Standard errors clustered by candidate city are reported in brackets. Standard errors clustered by department are reported in parentheses. Wild bootstrap p -values are reported in curly brackets. Controls include log population in 1793, log population in 1800 (except in column 1 of panel B), latitude and longitude, land area and land area squared, minimal and maximal altitude, a dummy for each of the four Ancien Régime administrative functions: *évêchés* (bishoprics), *bailliages* (bailiwicks), *recettes des finances* (tax centers), and *subdélégations*, distance to the department centroid, pre-reform market access, and wheat suitability in 20 km radius around each municipality. In column 6 of panel B, we look at the 1886 log population excluding civil servants and their families (see Section VD). In column 8, we lose some observations due to municipal mergers and splits taking place after 1914.

account for the growth of the civil service across the nineteenth century. Using data from Turquan (1899), we assume the following annual growth rates for the civil service across all departments: 1.42 percent before 1858, 1.86 percent between 1858 and 1873, and 1.04 percent from 1873 to 1911.⁴⁵ We then calculate time-varying,

⁴⁵ Historians have debated the accuracy of the data from Turquan (1899) and described the numerous challenges associated with enumerating civil servants in the nineteenth century (Rouban 2010; Ruiz 2016). Unfortunately, the data we could retrieve from statistical yearbooks in a subset of departments did not allow us to reconstruct better estimates of the evolution of the number of civil servants. We use the data from Turquan (1899) in this exercise but acknowledge its limitations.

department-level estimates of the civil servant population, applying these growth rates backward and forward starting from the 1886 census data, which, recall, contain exact estimates of the population of civil servants and their families. Finally, we allocate civil servants across capitals and other candidates following the same procedure as for 1886, and we residualize department fixed effects and baseline population. The two plots on the right-hand side of Figure 4 report the resulting time-varying estimates of the population net of civil servants and their families across both types of cities. These two figures display the same pattern, with a constant trend that breaks in the second half of the nineteenth century. Capitals eventually experience a demographic take-off relative to other candidate cities, but this take-off takes time to materialize, once the growth of the civil service is accounted for.

E. Contemporary Employment

While we emphasize the dynamic evolution of investments in state capacity over the subsequent century, it is also interesting to examine effects of the administrative reform in the very long term. In Tables 4 and 5 (final part of panel C), we use as dependent variables the 1999 municipal population as well as contemporary private sector employment observed between 2006 and 2015 (see Section IIIB).⁴⁶ More than two centuries after the reform, capitals host substantially more private (and public) sector jobs than other candidate cities.⁴⁷ Thus, the economic divergence that we observe in early twentieth century between capitals and other candidates clearly continued throughout the following century.

These differential patterns of growth over the two centuries following the reform are broadly consistent with the process described in Section II. Early investments in state capacity involve building up coercive power, including the ability to collect taxes and maintain law and order. Given these investments, the effects of state building on growth are limited in the short run. However, this eventually allows the state to collect more resources to fund local public goods. In the longer run, the higher availability of public goods in capitals makes these cities more attractive for citizens and firms alike. This leads to a long-term divergence between the winners and the losers of the administrative reform.

VI. Conclusion

One of the first missions entrusted to the members of France's first Constituent Assembly was to revamp the organization of the kingdom and to design a new administrative map reflecting modern principles of efficient government. This process involved choosing between century-old local centers and reallocating administrative functions to some cities at the expense of others. Anticipating that this reform would affect subsequent growth, the local elites exerted tremendous pressure on the

⁴⁶The 1999 estimate captures both natural growth and growth by absorption of neighboring municipalities after 1914.

⁴⁷We also find strong effects on public employment (results not reported). This effect is especially large for local public servants, that is, for the part of the public sector directly funded at the local level.

Assembly to ensure that their own cities would be chosen as the new local capitals. This lobbying effort was such that the Assembly, in artificial departments without a single dominant candidate, was unable to adjudicate between rival urban centers and even invented a rotation design where administrative functions would alternate across different cities.

In this paper, we show that the revolutionaries' expectations about the impacts of the reform were correct: differential investments in state capacity shaped patterns of economic development in the ensuing decades. In the long run, the cities chosen as the new local capitals ended up outgrowing other cities, and these capitals also drove faster growth in their periphery. However, we also show that this long-term divergence only materialized in the long run. In the immediate aftermath of the reform, citizens residing inside or near the new capitals were more exposed to investments in the coercive capacity of the state, including its ability to extract fiscal and physical resources through taxation and military conscription. Given the turmoil associated with the Napoleonic Wars until 1815, France's new local capitals paid a price in exchange for becoming the focal points of administrative presence.

Using a simple spatial equilibrium model, we show that investments in state capacity have dynamic as well as redistributive effects. Investments in the state's extractive and enforcement capacity must first be made before state building begins to deliver economic dividends. These investments then shape the spatial distribution of income and population over the long run. In the French case, despite multiple episodes of political upheaval during the nineteenth century, local investments in state capacity paid off and eventually contributed to the rise of the modern state. Nonetheless, the large political economy literature on institutions also shows that the dynamic sequence we observe between the initial administrative reform, investments in state capacity, and economic development may not have unfolded in the same way in other settings, especially if the initial buildup in coercive capacity is so large that it prevents the emergence of a cohesive society and a productive economy. Many other mediating factors, including mechanisms of rent extraction and social conflict, can put sand in the wheels of state building.

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